Number System

Topic Test of 100 Questions with Answers and Explanations · Class 10 Mathematics

What you will be tested on

- The kinds of numbers (natural, whole, integers, rational, irrational) and how they fit inside one another.
- Integers and signs, and comparing, simplifying and finding numbers between rational numbers.
- Ending and repeating decimals, and changing fractions to decimals and decimals to fractions.
- Square roots and cube roots, and telling rational numbers from irrational numbers.
- Brackets that contain square roots.
- Sums of natural, odd and even numbers, and problems on consecutive numbers.

Every question is based on a concept explained in the chapter document "Number System: Learn and Practise". Each question below is followed by its answer and explanation, and says where to revise it.

Test at a glance

Part	Topics covered	Questions
Part 1	Number families, integers and rational numbers: Kinds of numbers; integers and signs; simplify, compare and find between rational numbers	30
Part 2	Decimal expansions and conversions: Ending and repeating decimals; fractions to decimals; decimals to fractions	30
Part 3	Roots, irrational numbers and sums of natural numbers: Squares, cubes and roots; rational and irrational numbers; roots in brackets; sums and patterns of natural numbers	40
Total		**100**

- **Number of questions:** 100, all four-option multiple choice with exactly one correct option.
- **Suggested time:** about 117 minutes.
- **Marking scheme:** as per your teacher or the current board notification.

How to use this document

- **To take it as a test:** use a sheet of paper or a card to cover everything below each option list. Choose one option, write its letter, and then slide the card down to check.
- **To study from it:** read the question, try it, then read the answer and the steps. The line "How to see what this asks" shows how to start.
- **After a wrong answer:** read "Why the others are wrong" and then go to the chapter document using the "Revise in the chapter document" line.
- No calculators are needed; the numbers are small.

Difficulty analysis

Part	E (C / N)	M (C / N)	H (C / N)	Total
Part 1	16 (7 / 9)	12 (3 / 9)	2 (0 / 2)	30
Part 2	15 (5 / 10)	14 (3 / 11)	1 (0 / 1)	30
Part 3	20 (7 / 13)	16 (3 / 13)	4 (0 / 4)	40
All parts	**51 (19 / 32)**	**42 (9 / 33)**	**7 (0 / 7)**	**100**

- **Mix:** 51% easy, 42% medium and 7% hard (C = concept, N = calculation). Concept questions are 28 of 100 (28%).
- **Overall difficulty index:** 1.56 on a scale of 1 (all easy) to 3 (all hard).
- **Suggested time:** about 117 minutes (easy 45 s, medium 90 s, hard 135 s per question).
- **Level:** set to match the short, direct four-option questions of the reference papers: one or two lines, small clean numbers, mostly one concept and one to three steps. The calibration is for Mathematics and is based on the reference papers you supplied.

- **Answer letters:** a 25, b 25, c 25, d 25 (no letter more than 2 times in a row).

Basis in the chapter document

Every question rests on a concept explained in the chapter document. The table shows which type each question comes from.

Type in the chapter document	Questions	Count
Type 1: Which kind of number is it?	Q1, Q2, Q3, Q4, Q6, Q7, Q8, Q9, Q10	9
Type 2: Integers: order and signs	Q11, Q12, Q13, Q14, Q19, Q20, Q21, Q22, Q23, Q24, Q25, Q30	12
Type 3: Rational numbers: simplify, compare, find between	Q5, Q15, Q16, Q17, Q18, Q26, Q27, Q28, Q29	9
Type 4: Fractions to decimals: ending or repeating?	Q31, Q33, Q34, Q35, Q36, Q37, Q38, Q40, Q47, Q48, Q49, Q50, Q54, Q55, Q60	15
Type 5: Decimals to fractions	Q39, Q41, Q42, Q43, Q44, Q45, Q46, Q51, Q52, Q53, Q56, Q57, Q58, Q59	14
Type 6: Squares, cubes and their roots	Q67, Q70, Q71, Q72, Q73, Q74, Q75, Q76, Q77, Q86, Q87, Q88, Q89, Q90, Q91, Q92, Q93	17
Type 7: Irrational numbers	Q32, Q61, Q62, Q63, Q64, Q65, Q66, Q68, Q69, Q80, Q94	11
Type 8: Sums and patterns of natural numbers	Q81, Q82, Q83, Q84, Q85, Q98, Q99, Q100	8
Type 9: Roots inside brackets	Q78, Q79, Q95, Q96, Q97	5

Part 1 – Number families, integers and rational numbers

A. Concept questions

Q1 · Part 1 · Concept · Level E · MCQ

Q1. The smallest whole number is:

(a) 1 (b) −1
(c) 0 (d) There is no smallest whole number

Answer: (c) 0

How to see what this asks: This is a recall question about the family of whole numbers (W); see Type 1 in the chapter document.

Given: We need the smallest member of the whole numbers.

Rule: The whole numbers are 0, 1, 2, 3, …, the natural numbers together with 0.

Working: The list begins at 0 and goes upwards, and negative numbers are not whole numbers, so nothing comes before 0.

Result: The smallest whole number is 0.

Check: No whole number is smaller than 0 ✓.

Why the others are wrong: “1” is the smallest natural number, not the smallest whole number. “−1” is an integer but is not a whole number. “There is no smallest whole number” is false because the list has a first member, 0.

Concept behind the question: *The whole numbers W are the natural numbers together with 0 (Number families).*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 2

Q2 · Part 1 · Concept · Level E · MCQ

Q2. Which of the following is NOT an integer?

(a) −8 (b) 0
(c) 7/3 (d) 15

Answer: (c) 7/3

How to see what this asks: We need the option that has a fractional part; see Type 1.

Given: Four numbers; exactly one is not an integer.

Rule: An integer is a whole number or the negative of one; it has no fractional part.

Working: −8 is the negative of 8, 0 is a whole number and 15 is a whole number. But 7/3 = 2.333..., which lies between 2 and 3, so it is not a whole number or the negative of one.

Result: 7/3 is not an integer.

Check: 7 ÷ 3 leaves remainder 1, so 7/3 is not a whole number ✓.

Why the others are wrong: "−8" is a negative whole number, so it is an integer. "0" is an integer. "15" is a whole number, so it is an integer.

Concept behind the question: *Integers Z are the whole numbers and their negatives; a fraction like 7/3 is rational but not an integer.*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 3

Q3 · Part 1 · Concept · Level E · MCQ

Q3. Which of the following families contains −7 and 0 but does not contain 1/2?

(a) Integers
(b) Natural numbers
(c) Whole numbers
(d) Rational numbers

Answer: (a) Integers

How to see what this asks: We test each family against the three numbers; see Type 1.

Given: Three numbers: −7, 0 and 1/2.

Rule: N = 1, 2, 3, ...; W = 0, 1, 2, ...; Z = ..., −2, −1, 0, 1, 2, ...; Q = all numbers p/q.

Working: Natural numbers have neither −7 nor 0. Whole numbers have 0 but not −7. Integers have both −7 and 0, but 1/2 is not a whole number or its negative, so 1/2 is not in Z. Rational numbers contain all three, including 1/2.

Result: The family is the integers.

Check: Only Z contains −7 and 0 and leaves out 1/2 ✓.

Why the others are wrong: "Natural numbers" do not contain 0 or −7. "Whole numbers" do not contain −7. "Rational numbers" contain 1/2 as well.

Concept behind the question: *The families nest as N inside W inside Z inside Q; each new family adds new kinds of numbers.*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 1

Q4 · Part 1 · Concept · Level M · MCQ

Q4. Which of the following statements is correct?

(a) The difference of two natural numbers is always a natural number.
(b) The sum of two natural numbers is always a natural number.
(c) The quotient of two natural numbers is always a natural number.
(d) The difference of two whole numbers is always a whole number.

Answer: (b) The sum of two natural numbers is always a natural number.

How to see what this asks: We test each statement with an example that might break it.

Given: Four statements about adding, subtracting and dividing natural and whole numbers.

Rule: A statement that says "always" is false if one example breaks it.

Working: Sum: adding counting numbers always gives a counting number, for example 4 + 9 = 13, and no example breaks it. Difference of naturals: 3 − 8 = −5 is not natural. Quotient of naturals: 7 ÷ 2 = 3.5 is not natural. Difference of wholes: 2 − 6 = −4 is not whole.

Result: Only the statement about the sum of two natural numbers is always true.

Check: Each of the other three has a counter-example ✓.

Why the others are wrong: “The difference of two natural numbers is always a natural number.” fails for 3 − 8 = −5. “The quotient of two natural numbers is always a natural number.” fails for 7 ÷ 2 = 3.5. “The difference of two whole numbers is always a whole number.” fails for 2 − 6 = −4.

Concept behind the question: *Natural numbers are closed under addition (and multiplication) but not under subtraction or division.*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 2 · Rule R1 (a new family is needed when an operation breaks the old one)

Q5 · Part 1 · Concept · Level E · MCQ

Q5. Between any two different rational numbers there are:

(a) infinitely many rational numbers
(b) no other rational number
(c) exactly one rational number
(d) exactly ten rational numbers

Answer: (a) infinitely many rational numbers

How to see what this asks: This is a fact about the density of rational numbers; see Type 3.

Given: Two different rational numbers a and b.

Rule: The average (a + b) ÷ 2 of two rational numbers is rational and lies between them. Repeating the step gives more.

Working: Between 1/5 and 1/4 we can take their average. Then take the average of 1/5 and that average, and so on. The process never stops.

Result: There are infinitely many rational numbers between any two rational numbers.

Check: If there were only a fixed number n, taking one more average would give n + 1, which is a contradiction ✓.

Why the others are wrong: “no other rational number” is false because the average always lies between them. “exactly one rational number” is false because the process can be repeated. “exactly ten rational numbers” is false for the same reason: more can always be inserted.

Concept behind the question: *Rational numbers are "dense": there is always room between two of them.*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 9

Q6 · Part 1 · Concept · Level E · MCQ

Q6. Every integer n can be written in the form p/q (with q not zero) as:

(a) n/0
(b) 0/n
(c) 1/n
(d) n/1

Answer: (d) n/1

How to see what this asks: This is the link between integers and rational numbers; see Type 1.

Given: An integer n.

Rule: A rational number is a number written as p/q with q not zero.

Working: Dividing a number by 1 leaves it unchanged, so n = n/1, which has q = 1 and is allowed. The form n/0 is not allowed because the denominator is zero. Also 0/n equals 0, not n, and 1/n is the reciprocal of n, not n itself.

Result: n = n/1.

Check: For n = 7: 7/1 = 7 ✓.

Why the others are wrong: "n/0" is undefined because the denominator is zero. "0/n" equals 0 (for n not zero), not n. "1/n" is the reciprocal of n, not n itself.

Concept behind the question: *Every integer is a rational number because n = n/1.*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 2

Q7 · Part 1 · Concept · Level E · MCQ

Q7. Which of the following is a rational number that is also an integer?

(a) −6/3 (b) 5/2

(c) √7 (d) 0.25

Answer: (a) −6/3

How to see what this asks: Simplify each option first and then check the families; see Type 1.

Given: Four numbers. We need one that is both rational and an integer.

Rule: An integer is a whole number or the negative of one. A rational number can be written as p/q. Every integer is rational.

Working: −6/3 simplifies to −2, which is the negative of the whole number 2, so it is an integer (and also rational). 5/2 = 2.5 lies between 2 and 3, so it is not an integer. √7 is not the root of a perfect square (4 < 7 < 9), so it is irrational. 0.25 = 1/4 is rational but not an integer.

Result: −6/3 = −2.

Check: −2 × 3 = −6 ✓.

Why the others are wrong: "5/2" is rational but equals 2.5, which is not an integer. "√7" is irrational because 7 is not a perfect square. "0.25" is rational (1/4) but it is not an integer.

Concept behind the question: *A fraction can be an integer in disguise, so simplify before you decide (Rule R1 and Section 1.2).*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 3

Q8 · Part 1 · Concept · Level E · MCQ

Q8. Every point on the number line represents a:

(a) rational number (b) natural number

(c) real number (d) integer

Answer: (c) real number

How to see what this asks: This is the definition of the real numbers; see Section 1.2.

Given: The number line, with all its points.

Rule: The real numbers are all the rational numbers and all the irrational numbers together. Every point on the number line is a real number.

Working: Rational numbers fill many points, but a point such as √2 (the diagonal of a 1 cm square) is not rational. Irrational numbers also have points on the line. So only the real numbers cover every point.

Result: A real number.

Check: Both 3/4 (rational) and √2 (irrational) are points on the line, and both are real ✓.

Why the others are wrong: "rational number" misses points such as √2 and π. "natural number" covers only the points 1, 2, 3, ... "integer" covers only the whole-step points ..., −1, 0, 1, ...

Concept behind the question: *Real numbers = rational numbers + irrational numbers, and they fill the whole number line (Section 1.2, Rule R1).*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 1 · Section 1.2 (Real numbers (R))

Q9 · Part 1 · Concept · Level M · MCQ

Q9. Which of the following statements about the number families is true?

(a) A number can be both rational and irrational.
(b) Every irrational number is a real number.
(c) Every real number is irrational.
(d) Zero is an irrational number.

Answer: (b) Every irrational number is a real number.

How to see what this asks: Test each statement against the way the families fit together; see Type 1.

Given: Four statements; exactly one is true.

Rule: The real numbers are the rational and the irrational numbers together, and no number is both (Rule R1).

Working: Irrational numbers are one part of the real numbers, so the first statement is true. No number is both rational and irrational, so the second is false. The real number 5 is rational, so not every real number is irrational. Zero is 0/1, which is of the form p/q, so zero is rational.

Result: Every irrational number is a real number.

Check: The counter-example 5 breaks the third statement, and 0 = 0/1 breaks the fourth ✓.

Why the others are wrong: "A number can be both rational and irrational." is false because the two families never overlap. "Every real number is irrational." is false because 5 is real and rational. "Zero is an irrational number." is false because 0 = 0/1 is rational.

Concept behind the question: *Rational and irrational numbers are separate families that together make the real numbers (Rule R1).*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 2

Q10 · Part 1 · Concept · Level M · MCQ

Q10. The negative of a natural number is an integer, but it is never a:

(a) rational number
(b) real number
(c) number on the number line
(d) whole number

Answer: (d) whole number

How to see what this asks: Check which families contain negative whole numbers; see Type 1.

Given: A negative of a natural number, for example −3.

Rule: Whole numbers are 0, 1, 2, 3, … and have no negative members. Integers include the negative whole numbers. Every integer is rational and real (Rule R1).

Working: −3 is not in the list 0, 1, 2, 3, …, so it is not a whole number. It is an integer. It can be written −3/1, so it is rational. It is a real number. It also sits on the number line to the left of 0.

Result: A whole number.

Check: Whole numbers start at 0 and go up, so nothing negative can be whole ✓.

Why the others are wrong: "rational number" is true: −3 = −3/1. "real number" is true: every rational number is real. "number on the number line" is true: −3 is a point left of 0.

Concept behind the question: *The whole numbers contain no negative numbers; the integers are the family that adds them (Section 1.2, Rule R1).*

Revise in the chapter document: Type 1 (Which kind of number is it?) · Rule R1 · Example 2 · Section 1.2 (Integers)

B. Calculation questions

Q11 · Part 1 · Calculation · Level E · MCQ

Q11. The greatest of the numbers −12, −3, −20 and −7 is:

(a) −12 (b) −20

(c) −3 (d) −7

Answer: (c) −3

How to see what this asks: Ordering negatives on the number line; see Type 2.

Given: Four negative numbers; we want the greatest.

Rule: On the number line, the number nearest to 0 (to the right) is the greatest among negatives.

Working: The distances from 0 are 12, 3, 20 and 7. The smallest distance, 3, belongs to −3.

Result: The greatest number is −3.

Check: −20 < −12 < −7 < −3, so −3 is last in ascending order ✓.

Why the others are wrong: "−12" is farther from 0 than −3, so it is smaller. "−20" is the smallest of the four. "−7" is smaller than −3.

Concept behind the question: *With negative numbers, a bigger digit part means a smaller number.*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 4

Q12 · Part 1 · Calculation · Level E · MCQ

Q12. The value of (−7) × 6 is:

(a) 42 (b) −13

(c) −42 (d) 13

Answer: (c) −42

How to see what this asks: A single multiplication with different signs; see Type 2.

Given: (−7) × 6.

Rule: Different signs give a negative product: (negative) × (positive) = negative.

Working: 7 × 6 = 42, and because the signs differ the product is negative: −42.

Result: −42.

Check: Adding −7 six times gives −42 ✓.

Why the others are wrong: “42” ignores the sign rule. “−13” adds −7 and 6 instead of multiplying. “13” adds 7 and 6.

Concept behind the question: *Multiplication follows the sign rule: different signs give a negative result.*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 6

Q13 · Part 1 · Calculation · Level M · MCQ

Q13. The value of 24 ÷ (−4) − (−9) + 3 is:

(a) −12 (b) 18

(c) 0 (d) 6

Answer: (d) 6

How to see what this asks: Division first, then handle the double sign; see Type 2.

Given: 24 ÷ (−4) − (−9) + 3.

Rule: Divide first (different signs give a negative result). Then replace "− (−9)" by "+ 9" and work left to right.

Working: 24 ÷ (−4) = −6. Then −6 − (−9) = −6 + 9 = 3. Then 3 + 3 = 6.

Result: 6.

Check: Positives: 9 + 3 = 12; negatives: −6; 12 − 6 = 6 ✓.

Why the others are wrong: “−12” treats − (−9) as −9. “18” loses the negative sign in 24 ÷ (−4), giving 6, then adds 9 and 3. “0” adds −6 + 9 and then subtracts 3 instead of adding it.

Concept behind the question: *Subtracting a negative number is the same as adding the positive number.*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 5

Q14 · Part 1 · Calculation · Level M · MCQ

Q14. The value of (−5) × (−4) − 18 ÷ (−6) is:

(a) 17 (b) 23

(c) −23 (d) −17

Answer: (b) 23

How to see what this asks: Multiplication and division before subtraction, with a double sign at the end; see Type 2.

Given: (−5) × (−4) − 18 ÷ (−6).

Rule: Do the multiplication and the division first, using the sign rules; then subtract.

Working: (−5) × (−4) = +20, because the signs are the same. 18 ÷ (−6) = −3, because the signs differ. Then 20 − (−3) = 20 + 3 = 23.

Result: 23.

Check: The two products are +20 and −3; subtracting a negative number adds, so the answer is more than 20 ✓.

Why the others are wrong: “17” subtracts 3 instead of adding it, forgetting that 18 ÷ (−6) is negative. “−23” takes the product as −20 and then subtracts 3. “−17” takes the product as −20 and then adds 3.

Concept behind the question: *Settle the sign of each product or quotient before you add or subtract.*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 6

Q15 · Part 1 · Calculation · Level E · MCQ

Q15. 54/90 in its simplest form is:

(a) 27/45 (b) 9/15

(c) 3/5 (d) 6/10

Answer: (c) 3/5

How to see what this asks: Reduce by the greatest common factor; see Type 3.

Given: 54/90.

Rule: Divide the numerator and denominator by their greatest common factor.

Working: Both 54 and 90 have 18 as a factor, and no larger number divides both. So the greatest common factor is 18. Then 54 ÷ 18 = 3 and 90 ÷ 18 = 5.

Result: 3/5.

Check: 3 and 5 have no common factor, and 3 × 90 = 270 = 5 × 54 ✓.

Why the others are wrong: "27/45" is equal but can still be divided by 9. "9/15" is equal but can still be divided by 3. "6/10" is equal but can still be divided by 2.

Concept behind the question: *Simplest form means numerator and denominator share no factor except 1.*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 7

Q16 · Part 1 · Calculation · Level E · MCQ

Q16. Which is greater, 7/15 or 4/9?

(a) 4/9 (b) They are equal

(c) They cannot be compared (d) 7/15

Answer: (d) 7/15

How to see what this asks: Comparing two fractions by cross-multiplication; see Type 3.

Given: 7/15 and 4/9.

Rule: For a/b and c/d, compare a × d with c × b.

Working: 7 × 9 = 63 and 4 × 15 = 60. Since 63 > 60, the first fraction is greater.

Result: 7/15 is greater.

Check: Over 45: 7/15 = 21/45 and 4/9 = 20/45, and 21 > 20 ✓.

Why the others are wrong: "4/9" would need 4 × 15 to be larger than 7 × 9, but 60 < 63. "They are equal" is false because 63 is not equal to 60. "They cannot be compared" is false; any two rational numbers can be compared.

Concept behind the question: *Cross-multiplication compares two fractions without finding a common denominator.*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 8

Q17 · Part 1 · Calculation · Level M · MCQ

Q17. The smallest of 3/5, 5/8, 7/12 and 2/3 is:

(a) 3/5 (b) 5/8

(c) 7/12 (d) 2/3

Answer: (c) 7/12

How to see what this asks: Comparing four fractions over one common denominator; see Type 3.

Given: Four fractions with denominators 5, 8, 12 and 3.

Rule: Write all of them over a common denominator and compare the numerators.

Working: A common multiple of 5, 8, 12 and 3 is 120. Then 3/5 = 72/120, 5/8 = 75/120, 7/12 = 70/120 and 2/3 = 80/120. The smallest numerator is 70.

Result: The smallest fraction is 7/12.

Check: As decimals: 0.600, 0.625, 0.583 and 0.667; the smallest is 0.583 = 7/12 ✓.

Why the others are wrong: "3/5" is 72/120, bigger than 70/120. "5/8" is 75/120, bigger than 70/120. "2/3" is 80/120, the greatest of the four.

Concept behind the question: *A common denominator turns a comparison of fractions into a comparison of whole numbers.*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 10

Q18 · Part 1 · Calculation · Level M · MCQ

Q18. Which of the following lies between 1/5 and 1/4?

(a) 1/6 (b) 3/10

(c) 2/9 (d) 1/3

Answer: (c) 2/9

How to see what this asks: Placing a fraction between two others; see Type 3.

Given: The interval from 1/5 to 1/4, and four candidates.

Rule: Write the numbers over a common denominator (or as decimals) and see which candidate is in the interval.

Working: As decimals, 1/5 = 0.2 and 1/4 = 0.25. Then 2/9 = 0.222..., 1/6 = 0.167..., 3/10 = 0.3 and 1/3 = 0.333.... Only 0.222... is between 0.2 and 0.25.

Result: 2/9.

Check: Over 180: 1/5 = 36/180, 2/9 = 40/180 and 1/4 = 45/180, and 36 < 40 < 45 ✓.

Why the others are wrong: "1/6" is below 1/5. "3/10" is above 1/4. "1/3" is above 1/4.

Concept behind the question: *Decimals are a quick way to test whether a fraction lies between two others.*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 9

Q19 · Part 1 · Calculation · Level H · MCQ

Q19. The value of $7 - \{(-9) - (-4)\} \times (-3)$ is:

(a) 22 (b) −32

(c) −36 (d) −8

Answer: (d) −8

How to see what this asks: Brackets first, then multiplication, then subtraction; see Type 2.

Given: 7 − {(−9) − (−4)} × (−3).

Rule: Work inside the brackets first, then multiply, and subtract last.

Working: Inside the brackets: (−9) − (−4) = −9 + 4 = −5. Multiply: (−5) × (−3) = +15. Subtract: 7 − 15 = −8.

Result: −8.

Check: 7 − 15 is negative because 15 is greater than 7, and the difference is 8 ✓.

Why the others are wrong: "22" takes the product as −15 and then 7 − (−15) = 22. "−32" computes the brackets as −9 − 4 = −13, so the product is 39 and 7 − 39 = −32. "−36" subtracts first: (7 − (−5)) × (−3) = 12 × (−3).

Concept behind the question: *Brackets first; a negative times a negative is positive; subtract at the end.*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 6 · Rule R2 (brackets first)

Q20 · Part 1 · Calculation · Level M · MCQ

Q20. At noon the temperature was 6 °C. It fell by 4 °C every hour for 3 hours and then rose by 5 °C. The final temperature is:

(a) 7 °C
(b) −11 °C
(c) 11 °C
(d) −1 °C

Answer: (d) −1 °C

How to see what this asks: A story about ups and downs becomes integer addition; see Type 2.

Given: Start 6 °C; fall of 4 °C per hour for 3 hours; then a rise of 5 °C.

Rule: Final = start + total change; a fall is a negative change and a rise is positive.

Working: The total fall is 4 × 3 = 12, a change of −12. Then 6 + (−12) = −6. The rise of 5 gives −6 + 5 = −1.

Result: −1 °C.

Check: Total change = −12 + 5 = −7 and 6 − 7 = −1 ✓.

Why the others are wrong: "7 °C" uses only one hour of fall: 6 − 4 + 5. "−11 °C" subtracts the rise: 6 − 12 − 5. "11 °C" ignores the fall: 6 + 5.

Concept behind the question: *Treat falls as negative changes and add all the changes to the starting value.*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 5

Q21 · Part 1 · Calculation · Level E · MCQ

Q21. The ascending order (smallest first) of −7, 3, −10, 0 and 6 is:

(a) −10, −7, 0, 3, 6
(b) 6, 3, 0, −7, −10
(c) 0, 3, 6, −7, −10
(d) −7, −10, 0, 3, 6

Answer: (a) −10, −7, 0, 3, 6

How to see what this asks: Ordering integers on the number line; see Type 2.

Given: Five integers to arrange from smallest to greatest.

Rule: On the number line, numbers get bigger to the right. Every negative number is smaller than 0, and among negatives the one with the bigger digit part is smaller.

Working: The negatives are −7 and −10. Since 10 > 7, −10 is farther left, so −10 < −7. Then 0 comes next, then the positives 3 and 6.

Result: −10, −7, 0, 3, 6.

Check: Reading left to right, each number is bigger than the one before it ✓.

Why the others are wrong: "6, 3, 0, −7, −10" is the descending order. "0, 3, 6, −7, −10" orders by size ignoring the signs. "−7, −10, 0, 3, 6" puts the two negatives the wrong way round.

Concept behind the question: *With negative numbers, a bigger digit part means a smaller number (Type 2).*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 4

Q22 · Part 1 · Calculation · Level E · MCQ

Q22. The value of 15 − (−8) is:

(a) 7 (b) 23

(c) −23 (d) −7

Answer: (b) 23

How to see what this asks: Subtracting a negative number; see Type 2.

Given: 15 − (−8).

Rule: Subtracting a negative number is the same as adding the positive number: a − (−b) = a + b.

Working: 15 − (−8) = 15 + 8 = 23.

Result: 23.

Check: 23 + (−8) = 15, which brings us back to the start ✓.

Why the others are wrong: "7" subtracts 8 instead of adding it. "−23" gives the right size with the wrong sign. "−7" makes both the operation and the sign wrong.

Concept behind the question: *a − (−b) = a + b (Rule R2).*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 5

Q23 · Part 1 · Calculation · Level E · MCQ

Q23. The value of (−4) × (−5) × (−3) is:

(a) 60 (b) −12

(c) 12 (d) −60

Answer: (d) −60

How to see what this asks: Count the negative factors to settle the sign; see Type 2.

Given: (−4) × (−5) × (−3), with three negative factors.

Rule: Same signs give a positive product and different signs give a negative product. Multiply two numbers at a time.

Working: (−4) × (−5) = +20, because the signs are the same. Then 20 × (−3) = −60, because the signs differ.

Result: −60.

Check: 4 × 5 × 3 = 60, and an odd number of negative factors (three) gives a negative product ✓.

Why the others are wrong: "60" gets the size right but loses the sign; three negative factors give a negative product. "−12" adds the numbers instead of multiplying. "12" adds the numbers and also gets the sign wrong.

Concept behind the question: *An odd number of negative factors gives a negative product; an even number gives a positive one (Rule R2).*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 6

Q24 · Part 1 · Calculation · Level M · MCQ

Q24. The value of (−48) ÷ 6 × (−2) is:

(a) 16 (b) −16

(c) 4 (d) −4

Answer: (a) 16

How to see what this asks: Division and multiplication from left to right, with signs; see Type 2.

Given: (−48) ÷ 6 × (−2).

Rule: Do multiplication and division from left to right. Different signs give a negative result; same signs give a positive result.

Working: First (−48) ÷ 6 = −8, because the signs differ. Then (−8) × (−2) = +16, because the signs are the same.

Result: 16.

Check: 16 ÷ (−2) = −8 and (−8) × 6 = −48 ✓.

Why the others are wrong: "−16" loses the sign in the last step. "4" does 6 × (−2) = −12 first, then (−48) ÷ (−12) = 4. "−4" makes the order mistake and the sign mistake together.

Concept behind the question: *Work from left to right and settle the sign at every step (Rule R2).*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 6

Q25 · Part 1 · Calculation · Level M · MCQ

Q25. A lift is at floor 4 above the ground. It goes down 9 floors and then up 3 floors. The floor it is now at is:

(a) 2 (b) 10
(c) −2 (d) −8

Answer: (c) −2

How to see what this asks: A story about moves up and down becomes integer addition; see Type 2.

Given: Start at +4. Down 9 floors. Up 3 floors.

Rule: Going down is a negative change and going up is a positive change. Add all the changes to the start.

Working: 4 + (−9) + 3 = −5 + 3 = −2. The negative sign means 2 floors below the ground.

Result: −2.

Check: From −2, going down 3 gives −5, and going up 9 gives 4, which is the start ✓.

Why the others are wrong: "2" drops the sign, so it puts the lift above the ground. "10" treats going down as going up: 4 + 9 − 3. "−8" treats going up as going down: 4 − 9 − 3.

Concept behind the question: *Treat falls as negative changes and add all the changes to the starting value (Type 2).*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 5

Q26 · Part 1 · Calculation · Level E · MCQ

Q26. Which of the following fractions is NOT equal to 3/4?

(a) 9/12 (b) 15/20
(c) 18/25 (d) 21/28

Answer: (c) 18/25

How to see what this asks: Test each fraction by dividing top and bottom or by cross-multiplying; see Type 3.

Given: Four fractions; three equal 3/4 and one does not.

Rule: Dividing the top and the bottom by the same non-zero number gives an equal fraction. Two fractions a/b and c/d are equal when a × d = c × b.

Working: 9/12: divide by 3 to get 3/4. 15/20: divide by 5 to get 3/4. 21/28: divide by 7 to get 3/4. For 18/25: 18 × 4 = 72 but 3 × 25 = 75, and 72 ≠ 75.

Result: 18/25.

Check: 18/25 = 0.72 and 3/4 = 0.75, so they differ ✓.

Why the others are wrong: "9/12" simplifies to 3/4 (divide by 3). "15/20" simplifies to 3/4 (divide by 5). "21/28" simplifies to 3/4 (divide by 7).

Concept behind the question: *Equal fractions have the same simplest form, or equal cross-products (Rule R3).*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 7

Q27 · Part 1 · Calculation · Level M · MCQ

Q27. Which of the following fractions is greater than 5/7?

(a) 2/3 (b) 7/10

(c) 9/13 (d) 3/4

Answer: (d) 3/4

How to see what this asks: Compare each fraction with 5/7 by cross-multiplying; see Type 3.

Given: Compare four fractions with 5/7.

Rule: For a/b and c/d (b, d positive), compare a × d with c × b. The bigger product is on the side of the bigger fraction.

Working: 3/4 against 5/7: 3 × 7 = 21 and 5 × 4 = 20; 21 > 20, so 3/4 > 5/7. 2/3: 2 × 7 = 14 and 5 × 3 = 15; 14 < 15, so smaller. 7/10: 7 × 7 = 49 and 5 × 10 = 50; smaller. 9/13: 9 × 7 = 63 and 5 × 13 = 65; smaller.

Result: 3/4.

Check: As decimals: 5/7 ≈ 0.714 and 3/4 = 0.75, while 0.667, 0.7 and 0.692 are all below 0.714 ✓.

Why the others are wrong: "2/3" has cross-products 14 < 15, so it is smaller. "7/10" has cross-products 49 < 50, so it is smaller. "9/13" has cross-products 63 < 65, so it is smaller.

Concept behind the question: *Cross-multiplication compares two fractions without finding a common denominator (Rule R3).*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 8

Q28 · Part 1 · Calculation · Level M · MCQ

Q28. The ascending order of 4/5, 7/10 and 3/4 is:

(a) 7/10, 3/4, 4/5 (b) 3/4, 7/10, 4/5

(c) 4/5, 3/4, 7/10 (d) 7/10, 4/5, 3/4

Answer: (a) 7/10, 3/4, 4/5

How to see what this asks: Write the fractions over one common denominator; see Type 3.

Given: Three fractions to arrange from smallest to greatest.

Rule: Write all fractions over the same denominator. The biggest numerator is the biggest fraction.

Working: The denominators are 5, 10 and 4. A common multiple is 20. Then 4/5 = 16/20, 7/10 = 14/20 and 3/4 = 15/20. The numerators in order are 14, 15, 16.

Result: 7/10, 3/4, 4/5.

Check: As decimals: 0.7 < 0.75 < 0.8 ✓.

Why the others are wrong: "3/4, 7/10, 4/5" swaps the first two (15 and 14). "4/5, 3/4, 7/10" is the descending order. "7/10, 4/5, 3/4" swaps the last two (16 and 15).

Concept behind the question: *A common denominator turns a comparison of fractions into a comparison of whole numbers (Rule R3).*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 10

Q29 · Part 1 · Calculation · Level E · MCQ

Q29. A rational number lying between 5/11 and 6/11 is:

(a) 4/11 (b) 1/2

(c) 7/11 (d) 5/12

Answer: (b) 1/2

How to see what this asks: The tops 5 and 6 are next to each other, so double top and bottom first; see Type 3.

Given: The two fractions 5/11 and 6/11 have the same denominator and their tops have no gap.

Rule: Multiply the top and the bottom of both fractions by 2 to make room between the tops. Then pick a number between the new tops.

Working: 5/11 = 10/22 and 6/11 = 12/22. A number between 10/22 and 12/22 is 11/22, which equals 1/2. The other options: 4/11 is below 5/11 and 7/11 is above 6/11. Also 5/12 is below 5/11, because the same top over a bigger bottom is smaller.

Result: 1/2.

Check: As decimals: 0.4545 < 0.5 < 0.5455 ✓.

Why the others are wrong: "4/11" is smaller than 5/11. "7/11" is bigger than 6/11. "5/12" is smaller than 5/11 (bigger denominator, same top).

Concept behind the question: *When the tops have no gap, scale both fractions up to make room (Type 3).*

Revise in the chapter document: Type 3 (Rational numbers: simplify, compare, find between) · Rule R3 · Example 9

Q30 · Part 1 · Calculation · Level H · MCQ

Q30. The value of {(−3) × 8 − (−6)} ÷ (−2) is:

(a) 9 (b) −9

(c) 15 (d) 18

Answer: (a) 9

How to see what this asks: Brackets first, then multiplication, then subtraction, then division; see Type 2.

Given: {(−3) × 8 − (−6)} ÷ (−2).

Rule: Brackets come first. Inside, multiply before you subtract. Subtracting a negative adds the positive. Same signs give a positive quotient.

Working: Inside the bracket: (−3) × 8 = −24. Then −24 − (−6) = −24 + 6 = −18. Now divide: (−18) ÷ (−2) = +9.

Result: 9.

Check: 9 × (−2) = −18 and −18 = −24 + 6 ✓.

Why the others are wrong: "−9" loses the sign: −18 ÷ −2 is positive. "15" computes −24 − 6 = −30 inside the bracket, so −30 ÷ −2 = 15. "18" divides only the first term: (−24) ÷ (−2) + 6 = 18.

Concept behind the question: *Settle the bracket completely before you divide; the whole bracket is divided, not only its first term (Rule R2).*

Revise in the chapter document: Type 2 (Integers: order and signs) · Rule R2 · Example 6 · Rule R2 (brackets first)

Part 2 – Decimal expansions and conversions

A. Concept questions

Q31 · Part 2 · Concept · Level E · MCQ

Q31. Which of the following is a terminating decimal?

(a) 3/16 (b) 1/3

(c) 2/7 (d) 5/9

Answer: (a) 3/16

How to see what this asks: We test the denominator of each fraction; see Type 4.

Given: Four fractions in simplest form: 3/16, 1/3, 2/7 and 5/9.

Rule: A fraction in simplest form terminates only if its denominator has no prime factor other than 2 and 5.

Working: 16 = 2 × 2 × 2 × 2, only 2s, so 3/16 terminates (3 ÷ 16 = 0.1875). 3 is a prime other than 2 and 5, so 1/3 and 5/9 repeat. 7 is a prime other than 2 and 5, so 2/7 repeats.

Result: 3/16 is the terminating decimal.

Check: 3 ÷ 16 = 0.1875, which ends ✓.

Why the others are wrong: "1/3" has the prime 3 in the denominator: 0.333…. "2/7" has the prime 7 in the denominator, so it repeats. "5/9" has 3s in the denominator: 0.555….

Concept behind the question: *Only denominators made of 2s and 5s give terminating decimals.*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12

Q32 · Part 2 · Concept · Level E · MCQ

Q32. The decimal expansion of an irrational number is:

(a) terminating
(b) non-terminating and repeating
(c) non-terminating and non-repeating
(d) always a whole number

Answer: (c) non-terminating and non-repeating

How to see what this asks: This is the definition of an irrational number in terms of its decimal; see Type 7.

Given: A number that is irrational, meaning it cannot be written as p/q.

Rule: Rational numbers have decimals that end or repeat; so an irrational number must have a decimal that does neither.

Working: If its decimal ended or repeated, we could turn it into a fraction p/q (Type 5), so it would be rational. Hence the decimal of an irrational number neither ends nor repeats.

Result: Non-terminating and non-repeating.

Check: Example: √2 = 1.41421356… has no end and no repeating block ✓.

Why the others are wrong: "terminating" would make the number rational. "non-terminating and repeating" would also make the number rational. "always a whole number" whole numbers are rational.

Concept behind the question: *Rational: decimal ends or repeats. Irrational: neither.*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 25

Q33 · Part 2 · Concept · Level E · MCQ

Q33. A fraction p/q in simplest form has a terminating decimal expansion if the prime factors of q are only:

(a) 2 and/or 5
(b) 3 and/or 7
(c) 2, 3 or 5
(d) any primes

Answer: (a) 2 and/or 5

How to see what this asks: This is the terminating test; see Type 4.

Given: A fraction p/q with no common factor in p and q.

Rule: Only primes that divide 10 (which are 2 and 5) can make the denominator a power of 10.

Working: If q = 2^m × 5^n, multiply top and bottom by enough 2s or 5s to make the denominator 10^k. Then the decimal ends after k places. A prime such as 3 can never be turned into a power of 10, so the decimal repeats.

Result: The prime factors of q must be only 2 and/or 5.

Check: Example: 7/40, 40 = 2^3 × 5, and 7/40 = 0.175 ends ✓.

Why the others are wrong: "3 and/or 7" these primes make the decimal repeat. "2, 3 or 5" the prime 3 would make the decimal repeat. "any primes" is false: only 2 and 5 are allowed.

Concept behind the question: *Terminating decimals come from denominators of the form 2^m × 5^n.*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12

Q34 · Part 2 · Concept · Level M · MCQ

Q34. Which of the following is a non-terminating repeating decimal?

(a) 0.125 (b) 0.1010010001…

(c) 0.454545… (d) 2.5

Answer: (c) 0.454545…

How to see what this asks: We separate three kinds of decimal: ending, repeating and neither; see Type 4 and Type 7.

Given: Four decimals.

Rule: Terminating: ends. Repeating: goes on forever with a block that repeats. Neither: goes on forever with no repeating block.

Working: 0.125 and 2.5 end. 0.1010010001… goes on forever, but the number of zeros between the 1s keeps growing, so no block repeats. 0.454545… goes on forever with the block 45 repeating.

Result: 0.454545…

Check: 0.4545… = 45/99 = 5/11, a fraction ✓.

Why the others are wrong: “0.125” ends after three places. “0.1010010001…” is non-terminating but non-repeating (irrational). “2.5” ends after one place.

Concept behind the question: *A repeating decimal is rational; a decimal that never ends and never repeats is irrational.*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 11 · Section 1.2 (Decimal expansion)

Q35 · Part 2 · Concept · Level M · MCQ

Q35. Which of the following fractions has a non-terminating repeating decimal?

(a) 14/66 (b) 39/130

(c) 27/90 (d) 18/45

Answer: (a) 14/66

How to see what this asks: Reduce each fraction first, then test the denominator; see Type 4.

Given: Four fractions that are not all in simplest form.

Rule: Reduce to simplest form; the decimal terminates only if the denominator then has just the primes 2 and 5.

Working: 39/130 = 3/10 (divide by 13), and 10 = 2 × 5, so it terminates. 27/90 = 3/10 (divide by 9), so it terminates. 18/45 = 2/5 (divide by 9), so it terminates. 14/66 = 7/33 (divide by 2), and 33 = 3 × 11 has primes other than 2 and 5.

Result: 14/66 (= 7/33) repeats.

Check: 7 ÷ 33 = 0.212121…, a repeating block ✓.

Why the others are wrong: “39/130” reduces to 3/10 = 0.3. “27/90” reduces to 3/10 = 0.3. “18/45” reduces to 2/5 = 0.4.

Concept behind the question: *Always reduce first: a factor that looks "bad" in the denominator may cancel.*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 13

Q36 · Part 2 · Concept · Level E · MCQ

Q36. The decimal expansion of 1/7 is:

(a) terminating (b) non-terminating and repeating

(c) non-terminating and non-repeating (d) a whole number

Answer: (b) non-terminating and repeating

How to see what this asks: Apply the terminating test to the denominator; see Type 4.

Given: The fraction 1/7, already in simplest form.

Rule: A fraction in simplest form ends only if its denominator has no prime factor except 2 and 5. Any other prime makes the decimal repeat.

Working: The denominator 7 is a prime other than 2 and 5, so the decimal does not end. A rational number never goes on without a pattern, so it repeats.

Result: Non-terminating and repeating.

Check: Long division gives 1/7 = 0.142857142857..., and the block 142857 repeats ✓.

Why the others are wrong: "terminating" needs a denominator of only 2s and 5s, and 7 is neither. "non-terminating and non-repeating" describes irrational numbers, and 1/7 is rational. "a whole number" is wrong because 1/7 is less than 1.

Concept behind the question: *The prime 7 in the denominator forces a repeating decimal (Rule R4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12

Q37 · Part 2 · Concept · Level E · MCQ

Q37. The decimal expansion of every rational number is either terminating or:

(a) non-terminating and non-repeating
(b) a negative number
(c) non-terminating and repeating
(d) a whole number

Answer: (c) non-terminating and repeating

How to see what this asks: This is the link between rational numbers and their decimals; see Section 1.2.

Given: Any rational number p/q.

Rule: The decimal of a rational number either ends or goes on forever with a block of digits repeating.

Working: Long division by q can leave only q − 1 different non-zero remainders. So either a remainder is 0 (the decimal ends) or a remainder comes back (the digits repeat).

Result: Non-terminating and repeating.

Check: 1/4 = 0.25 ends and 1/3 = 0.333... repeats ✓.

Why the others are wrong: "non-terminating and non-repeating" is the decimal of an irrational number. "a negative number" confuses the sign of a number with the shape of its decimal. "a whole number" is true only for a few rational numbers such as 6/3.

Concept behind the question: *Rational: the decimal ends or repeats. Irrational: it does neither (Section 1.2, Rule R4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 11 · Section 1.2 (Decimal expansion)

Q38 · Part 2 · Concept · Level M · MCQ

Q38. In simplest form, a fraction has denominator $2^3 \times 5^2 \times 7$. Its decimal expansion is:

(a) terminating after 3 places
(b) terminating after 5 places
(c) non-terminating and non-repeating
(d) non-terminating and repeating

Answer: (d) non-terminating and repeating

How to see what this asks: Look for a prime other than 2 and 5 in the denominator; see Type 4.

Given: The denominator $2^3 \times 5^2 \times 7$, with the fraction in simplest form.

Rule: The decimal ends only if the denominator has no prime factor except 2 and 5.

Working: The denominator contains the prime 7, so the decimal cannot end. It is a rational number, so its digits repeat.

Result: Non-terminating and repeating.

Check: Without the factor 7 the denominator $2^3 \times 5^2$ would give 3 decimal places, but the 7 spoils it ✓.

Why the others are wrong: "terminating after 3 places" would be true only if the factor 7 were absent (the larger power is 3). "terminating after 5 places" adds the powers 3 + 2, which is not the rule, and ignores the 7. "non-terminating and non-repeating" is impossible for a rational number.

Concept behind the question: *One prime other than 2 or 5 in the simplest-form denominator makes the decimal repeat (Rule R4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12

B. Calculation questions

Q39 · Part 2 · Calculation · Level E · MCQ

Q39. 0.35 written as a fraction in simplest form is:

(a) 7/20 (b) 35/10
(c) 3/5 (d) 7/10

Answer: (a) 7/20

How to see what this asks: Place value, then reduce; see Type 5.

Given: The decimal 0.35, with two places.

Rule: Write the digits over 100, then divide top and bottom by the greatest common factor.

Working: 0.35 = 35/100. The greatest common factor of 35 and 100 is 5, so 35 ÷ 5 = 7 and 100 ÷ 5 = 20.

Result: 7/20.

Check: 7 ÷ 20 = 0.35 ✓.

Why the others are wrong: "35/10" equals 3.5. "3/5" equals 0.6. "7/10" equals 0.7.

Concept behind the question: *Two decimal places mean a denominator of 100 before reducing.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 15

Q40 · Part 2 · Calculation · Level E · MCQ

Q40. 7/25 written as a decimal is:

(a) 0.7 (b) 0.25
(c) 2.8 (d) 0.28

Answer: (d) 0.28

How to see what this asks: Turn the denominator into 100 by multiplying; see Type 4.

Given: 7/25.

Rule: Multiply top and bottom by the number that makes the denominator 100 (or divide 7 by 25).

Working: 25 × 4 = 100, so 7/25 = (7 × 4)/(25 × 4) = 28/100 = 0.28.

Result: 0.28.

Check: 25 × 0.28 = 7 ✓.

Why the others are wrong: "0.7" divides 7 by 10 instead of 25. "0.25" uses the denominator as the decimal. "2.8" puts the decimal point one place wrong.

Concept behind the question: *If the denominator divides a power of 10, scale the fraction to that power of 10.*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12

Q41 · Part 2 · Calculation · Level E · MCQ

Q41. The value of 0.444… + 0.222… as a fraction is:

(a) 3/5 (b) 6/99

(c) 2/3 (d) 1/6

Answer: (c) 2/3

How to see what this asks: Convert each repeating decimal to a fraction, then add; see Type 5.

Given: 0.444… and 0.222…, each with one repeating digit.

Rule: A single repeating digit d over 9: 0.ddd… = d/9.

Working: 0.444… = 4/9 and 0.222… = 2/9. Their sum is 4/9 + 2/9 = 6/9 = 2/3.

Result: 2/3.

Check: 2 ÷ 3 = 0.666…, and 0.444… + 0.222… = 0.666… ✓.

Why the others are wrong: “3/5” equals 0.6, which ends. “6/99” uses 99, which is for two repeating digits. “1/6” is 0.1666….

Concept behind the question: *One repeating digit d means d/9; the fractions then add as usual.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 19

Q42 · Part 2 · Calculation · Level M · MCQ

Q42. If 0.363636… = p/q in simplest form, then p + q is:

(a) 15 (b) 135

(c) 34 (d) 13

Answer: (a) 15

How to see what this asks: Convert the repeating decimal, then add the top and bottom; see Type 5.

Given: 0.363636…, where the block 36 repeats.

Rule: Let x = 0.3636…, multiply by 100 and subtract.

Working: 100x = 36.3636…, so 100x − x = 36 and 99x = 36, giving x = 36/99. The common factor is 9, so x = 4/11. Then p + q = 4 + 11 = 15.

Result: 15.

Check: 4 ÷ 11 = 0.3636… ✓.

Why the others are wrong: “135” adds the unreduced 36 and 99. “34” uses 36/100 = 9/25 and adds 9 + 25. “13” uses a denominator of 9 for one repeating digit: 4/9.

Concept behind the question: *A block of two digits goes over 99; reduce before adding p and q.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 17

Q43 · Part 2 · Calculation · Level M · MCQ

Q43. 0.666… × 0.333… expressed as a fraction is:

(a) 9/50 (b) 2/9

(c) 2/11 (d) 1/3

Answer: (b) 2/9

How to see what this asks: Convert both repeating decimals to fractions first, then multiply; see Type 5.

Given: 0.666… and 0.333….

Rule: A single repeating digit d is d/9.

Working: 0.666… = 6/9 = 2/3 and 0.333… = 3/9 = 1/3. The product is (2/3) × (1/3) = 2/9.

Result: 2/9.

Check: 2 ÷ 9 = 0.222…, and 0.666… × 0.333… ≈ 0.2222 ✓.

Why the others are wrong: "9/50" multiplies 0.6 × 0.3 = 0.18, treating the decimals as ending. "2/11" multiplies 6 × 3 and puts it over 99. "1/3" is just the second number.

Concept behind the question: *Convert repeating decimals to fractions before you do arithmetic with them.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 19 · Rule R5 (multiply as fractions)

Q44 · Part 2 · Calculation · Level M · MCQ

Q44. 1.222… (the digit 2 repeats) expressed as a fraction is:

(a) 11/9 (b) 6/5

(c) 4/3 (d) 122/99

Answer: (a) 11/9

How to see what this asks: Split off the whole part and convert the repeating part; see Type 5.

Given: 1.222…, with a whole part 1 and a repeating digit 2.

Rule: Write the number as whole part + repeating part, and 0.222… = 2/9.

Working: 1.222… = 1 + 0.222… = 1 + 2/9 = 9/9 + 2/9 = 11/9.

Result: 11/9.

Check: 11 ÷ 9 = 1.2222… ✓.

Why the others are wrong: "6/5" is 1.2, which ends. "4/3" is 12/9, which is 1.333…. "122/99" is 1.2323…, a different repeating decimal.

Concept behind the question: *The whole part stays; only the repeating part needs the subtraction trick.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 16 · Rule R5 (a whole part stays)

Q45 · Part 2 · Calculation · Level H · MCQ

Q45. The value of 0.1212… + 0.2121… is:

(a) 33/100 (b) 11/30

(c) 3/10 (d) 1/3

Answer: (d) 1/3

How to see what this asks: Convert both two-digit repeating decimals into fractions over 99; see Type 5.

Given: 0.1212… (block 12) and 0.2121… (block 21).

Rule: A repeating block of two digits over 99.

Working: 0.1212… = 12/99 and 0.2121… = 21/99. The sum is (12 + 21)/99 = 33/99 = 1/3.

Result: 1/3.

Check: 0.1212… + 0.2121… = 0.3333…, which is 1/3 ✓.

Why the others are wrong: "33/100" uses 100 as the denominator, which gives 0.33 exactly. "11/30" is 0.3666…. "3/10" is 0.3, which ends.

Concept behind the question: *Repeating blocks add like fractions with the same denominator 99.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 19

Q46 · Part 2 · Calculation · Level E · MCQ

Q46. Which fraction is equal to 0.04?

(a) 1/4 (b) 4/10
(c) 1/40 (d) 1/25

Answer: (d) 1/25

How to see what this asks: Place value, then reduce; see Type 5.

Given: 0.04, which has two decimal places.

Rule: Write the digits over 100: 0.04 = 4/100, then reduce.

Working: 4/100 divided top and bottom by 4 is 1/25.

Result: 1/25.

Check: 1 ÷ 25 = 0.04 ✓.

Why the others are wrong: "1/4" equals 0.25. "4/10" equals 0.4. "1/40" equals 0.025.

Concept behind the question: *A zero just after the decimal point makes the fraction smaller; keep track of the place values.*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 15

Q47 · Part 2 · Calculation · Level M · MCQ

Q47. The decimal expansion of 17/125 terminates after how many places of decimals?

(a) 2 (b) 4
(c) 3 (d) 5

Answer: (c) 3

How to see what this asks: The denominator 125 is a power of 5, which tells us how many places; see Type 4.

Given: 17/125, already in simplest form.

Rule: Write the denominator as 2^m × 5^n; the decimal ends after the larger of m and n places.

Working: 125 = 5 × 5 × 5 = 5^3, so m = 0 and n = 3. To make the denominator a power of 10 we multiply by $2^3 = 8$: 17/125 = (17 × 8)/(125 × 8) = 136/1000 = 0.136.

Result: 3 places.

Check: 0.136 has three digits after the decimal point ✓.

Why the others are wrong: "2" would be right for a denominator of 100. "4" would be right for a denominator of 10000. "5" confuses the power 3 with the factor 5.

Concept behind the question: *The number of decimal places is the larger power of 2 or 5 in the denominator.*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12 · Rule R4 (number of decimal places)

Q48 · Part 2 · Calculation · Level E · MCQ

Q48. The decimal form of 3/8 is:

(a) 0.375 (b) 0.38
(c) 0.125 (d) 0.35

Answer: (a) 0.375

How to see what this asks: Divide the top by the bottom; see Type 4.

Given: The fraction 3/8.

Rule: A fraction p/q means p ÷ q. Write 3 as 3.000… and divide, bringing down a zero each time.

Working: 30 ÷ 8 = 3 remainder 6. Then 60 ÷ 8 = 7 remainder 4. Then 40 ÷ 8 = 5 remainder 0. The remainder is 0, so the division stops: 0.375.

Result: 0.375.

Check: 8 × 0.375 = 3 ✓.

Why the others are wrong: "0.38" is rounded, not exact. "0.125" is the decimal of 1/8. "0.35" mixes up the digits of the division.

Concept behind the question: *Long division ends when a remainder is 0; that gives a terminating decimal (Type 4, Example 11).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 11

Q49 · Part 2 · Calculation · Level E · MCQ

Q49. The decimal form of 9/20 is:

(a) 0.54 (b) 4.5
(c) 0.9 (d) 0.45

Answer: (d) 0.45

How to see what this asks: Turn the denominator into 100, or divide 9 by 20; see Type 4.

Given: The fraction 9/20.

Rule: Multiply the top and the bottom by the same number to get a denominator of 10, 100 or 1000.

Working: 20 × 5 = 100, so multiply the top and the bottom by 5: 9/20 = 45/100 = 0.45.

Result: 0.45.

Check: 20 × 0.45 = 9 ✓.

Why the others are wrong: "0.54" reverses the digits. "4.5" puts the decimal point in the wrong place. "0.9" is 9/10 and ignores the 2 in the denominator.

Concept behind the question: *A denominator made of 2s and 5s can be scaled to a power of 10 (Rule R4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 12

Q50 · Part 2 · Calculation · Level E · MCQ

Q50. The decimal form of 5/11 is:

(a) 0.4545… (the block 45 repeats) (b) 0.5454… (the block 54 repeats)
(c) 0.4555… (only the 5 repeats) (d) 0.45 (it ends)

Answer: (a) 0.4545… (the block 45 repeats)

How to see what this asks: Divide 5 by 11 and watch for a remainder that comes back; see Type 4.

Given: The fraction 5/11.

Rule: Divide the top by the bottom. If a remainder repeats, the digits repeat from there.

Working: 50 ÷ 11 = 4 remainder 6. Then 60 ÷ 11 = 5 remainder 5. Then 50 ÷ 11 = 4 remainder 6 again. The remainder 6 and the remainder 5 now repeat, so the block 45 repeats.

Result: 0.4545… with the block 45 repeating.

Check: 11 × 0.4545… = 4.9999… = 5 ✓.

Why the others are wrong: "0.5454… (the block 54 repeats)" starts with the wrong digit; the first digit is 4. "0.4555… (only the 5 repeats)" repeats only one digit, but the remainders show a block of two. "0.45 (it ends)" is impossible because the prime 11 is in the denominator.

Concept behind the question: *A remainder that comes back means the digits repeat (Type 4, Example 11).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 11

Q51 · Part 2 · Calculation · Level E · MCQ

Q51. Express 0.625 as a fraction in simplest form:

(a) 25/40 (b) 8/5

(c) 5/8 (d) 31/50

Answer: (c) 5/8

How to see what this asks: Place value, then simplify; see Type 5.

Given: The decimal 0.625, which has three digits after the point.

Rule: A decimal with k digits after the point is its digits over 1 followed by k zeros. Then divide the top and the bottom by their highest common factor.

Working: 0.625 = 625/1000. Both end in 5 or 0, so divide by 5: 125/200. Divide by 5 again: 25/40. Divide by 5 again: 5/8. Now 5 is prime and does not divide 8, so we stop.

Result: 5/8.

Check: 5 ÷ 8 = 0.625 ✓.

Why the others are wrong: "25/40" is not in simplest form: 5 divides both numbers. "8/5" is the reciprocal. "31/50" is 0.62 and drops the last digit 5.

Concept behind the question: *Three decimal places mean thousandths; simplify until no common factor is left (Rule R5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 15

Q52 · Part 2 · Calculation · Level E · MCQ

Q52. Express 0.151515… (the block 15 repeats) as a fraction in simplest form:

(a) 3/20 (b) 5/33

(c) 1/6 (d) 5/3

Answer: (b) 5/33

How to see what this asks: Two digits repeat, so multiply by 100 and subtract; see Type 5.

Given: x = 0.151515…, with the block 15 repeating.

Rule: Two digits repeat, so multiply by 100. Subtract x so the endless tails cancel.

Working: x = 0.1515… (line A). 100x = 15.1515… (line B). Subtract: 100x − x = 15, so 99x = 15 and x = 15/99. The highest common factor of 15 and 99 is 3, so x = 5/33.

Result: 5/33.

Check: 5 ÷ 33: 50 ÷ 33 = 1 remainder 17; 170 ÷ 33 = 5 remainder 5; then 50 again, giving 0.1515… ✓.

Why the others are wrong: "3/20" is 15/100 = 0.15, which ends. "1/6" comes from 15/90, which is the denominator for a mixed decimal. "5/3" flips the fraction and uses 9 instead of 99.

Concept behind the question: *A repeating block of two digits is the block over 99 (Rule R5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 17

Q53 · Part 2 · Calculation · Level M · MCQ

Q53. Express 0.8333… (only the digit 3 repeats) as a fraction in simplest form:

(a) 83/99 (b) 83/100

(c) 4/5 (d) 5/6

Answer: (d) 5/6

How to see what this asks: A mixed recurring decimal: use 10x and 100x; see Type 5.

Given: x = 0.8333…. The 8 does not repeat; the 3 does.

Rule: Make one line with the point just before the repeating 3s and another with the point just after the first 3. Then subtract.

Working: 10x = 8.333… and 100x = 83.333…. Subtract: 100x − 10x = 83.333… − 8.333…, so 90x = 75. Then x = 75/90. Dividing the top and the bottom by 15 gives 5/6.

Result: 5/6.

Check: 5 ÷ 6 = 0.8333… ✓.

Why the others are wrong: "83/99" treats both digits as repeating (0.8383…). "83/100" treats the decimal as ending at 0.83. "4/5" is 0.8, which ignores the repeating 3s.

Concept behind the question: *In a mixed recurring decimal, the non-repeating digits stay outside the repeat (Rule R5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 18

Q54 · Part 2 · Calculation · Level E · MCQ

Q54. The decimal form of 7/16 is:

(a) 0.4357 (b) 0.4375

(c) 0.3475 (d) 0.875

Answer: (b) 0.4375

How to see what this asks: Divide 7 by 16; see Type 4.

Given: The fraction 7/16.

Rule: Divide the top by the bottom, bringing down a zero each time.

Working: 70 ÷ 16 = 4 remainder 6. Then 60 ÷ 16 = 3 remainder 12. Then 120 ÷ 16 = 7 remainder 8. Then 80 ÷ 16 = 5 remainder 0. So the decimal is 0.4375.

Result: 0.4375.

Check: 16 × 0.4375 = 7 ✓.

Why the others are wrong: "0.4357" swaps the last two digits. "0.3475" has the wrong digits from the division. "0.875" is the decimal of 7/8.

Concept behind the question: *Dividing until the remainder is 0 gives a terminating decimal (Type 4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 11

Q55 · Part 2 · Calculation · Level M · MCQ

Q55. Which of the following fractions has a terminating decimal expansion?

(a) 21/45 (b) 14/21

(c) 5/12 (d) 33/66

Answer: (d) 33/66

How to see what this asks: Reduce each fraction before you look at its denominator; see Type 4.

Given: Four fractions; only one ends.

Rule: Reduce to simplest form first. Then the decimal ends only if the denominator has no prime factor except 2 and 5.

Working: 33/66 = 1/2 (divide by 33), and the denominator 2 is allowed, so 0.5 ends. 21/45 = 7/15 and 15 = 3 × 5 contains 3, so it repeats. 14/21 = 2/3 and the 3 makes it repeat. 5/12 is already simplest and 12 = 2 × 2 × 3 contains 3, so it repeats.

Result: 33/66.

Check: 33/66 = 0.5, while 7/15 = 0.4666..., 2/3 = 0.666... and 5/12 = 0.41666... ✓.

Why the others are wrong: "21/45" reduces to 7/15, and the 3 in 15 makes it repeat. "14/21" reduces to 2/3, which repeats. "5/12" has the prime 3 in 12, so it repeats.

Concept behind the question: *Always reduce first: a bad prime in the denominator may cancel (Rule R4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 13

Q56 · Part 2 · Calculation · Level M · MCQ

Q56. If 0.1818... = p/q in simplest form, then q − p is:

(a) 9 (b) 13

(c) 81 (d) 11

Answer: (a) 9

How to see what this asks: Convert the repeating decimal, reduce it, then subtract; see Type 5.

Given: 0.1818..., with the block 18 repeating, written as p/q in simplest form.

Rule: A block of two digits is the block over 99. Reduce before reading p and q.

Working: 0.1818... = 18/99. The highest common factor of 18 and 99 is 9, so 18/99 = 2/11. So p = 2 and q = 11. Then q − p = 11 − 2 = 9.

Result: 9.

Check: 2 ÷ 11 = 0.1818... ✓.

Why the others are wrong: "13" is p + q, not q − p. "81" is 99 − 18, taken before reducing. "11" is q alone.

Concept behind the question: *Reduce the fraction first, then read p and q (Rule R5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 17

Q57 · Part 2 · Calculation · Level M · MCQ

Q57. The value of 0.8888... − 0.3333... as a fraction is:

(a) 1/2 (b) 5/9

(c) 5/99 (d) 11/9

Answer: (b) 5/9

How to see what this asks: Convert each repeating decimal into a fraction, then subtract; see Type 5.

Given: 0.8888... − 0.3333..., with one repeating digit in each.

Rule: One repeating digit is that digit over 9. Subtract the fractions over the same denominator.

Working: 0.8888... = 8/9 and 0.3333... = 3/9. So 8/9 − 3/9 = 5/9.

Result: 5/9.

Check: 0.8888... − 0.3333... = 0.5555... and 5/9 = 0.5555... ✓.

Why the others are wrong: "1/2" reads the answer 0.5555... as 0.5, which ends. "5/99" uses 99, which is for a block of two digits. "11/9" adds the fractions instead of subtracting.

Concept behind the question: *Turn repeating decimals into fractions before doing arithmetic with them (Rule R5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 19

Q58 · Part 2 · Calculation · Level M · MCQ

Q58. Express 3.1818… (the block 18 repeats) as a fraction:

(a) 159/50 (b) 106/33

(c) 35/11 (d) 31/9

Answer: (c) 35/11

How to see what this asks: Keep the whole part and convert only the repeating part; see Type 5.

Given: 3.1818…, with a whole part 3 and the block 18 repeating.

Rule: A whole part stays as it is. A repeating block of two digits is the block over 99.

Working: 3.1818… = 3 + 0.1818…. Now 0.1818… = 18/99 = 2/11. So the number is 3 + 2/11 = 33/11 + 2/11 = 35/11.

Result: 35/11.

Check: 35 ÷ 11 = 3 remainder 2; 20 ÷ 11 = 1 remainder 9; 90 ÷ 11 = 8 remainder 2, so 3.1818… ✓.

Why the others are wrong: “159/50” is 3.18, which ends. “106/33” is 318/99, which is 3.2121…. “31/9” is 3.444…, which is not the same repeat.

Concept behind the question: *Split off the whole part, convert the repeating part, then add (Rule R5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 17 · Rule R5 (a whole part stays)

Q59 · Part 2 · Calculation · Level M · MCQ

Q59. The ascending order of 0.4444…, 0.45 and 0.4545… is:

(a) 0.4444…, 0.45, 0.4545… (b) 0.45, 0.4444…, 0.4545…

(c) 0.4545…, 0.45, 0.4444… (d) 0.4444…, 0.4545…, 0.45

Answer: (a) 0.4444…, 0.45, 0.4545…

How to see what this asks: Write the three decimals with the same number of digits and compare them; see Type 5 and Type 3.

Given: Three decimals, two of them repeating.

Rule: Compare digit by digit from the left, writing the repeating decimals out far enough.

Working: 0.4444… = 0.44444…, 0.45 = 0.45000…, 0.4545… = 0.45454…. The first digit is 4 in all three. The second digit is 4, 5, 5. So 0.4444… is smallest. For the other two the third digit is 0 and 4, so 0.45000 is smaller than 0.45454.

Result: 0.4444…, 0.45, 0.4545….

Check: As fractions: 4/9 = 0.4444…, 9/20 = 0.45 and 15/33 = 5/11 = 0.4545…, and 40/90 < 40.5/90 < 40.9/90 ✓.

Why the others are wrong: “0.45, 0.4444…, 0.4545…” puts 0.4444… in the middle, but its second digit 4 is less than 5. “0.4545…, 0.45, 0.4444…” is the descending order. “0.4444…, 0.4545…, 0.45” thinks 0.4545… is smaller than 0.45, but its third digit 4 is more than 0.

Concept behind the question: *A repeating decimal must be written out far enough to compare it with a terminating one (Type 5).*

Revise in the chapter document: Type 5 (Decimals to fractions) · Rule R5 · Example 18 · Type 3 (comparing)

Q60 · Part 2 · Calculation · Level M · MCQ

Q60. The decimal form of 11/12 is:

(a) 0.9111… (only the 1 repeats) (b) 0.9333… (only the 3 repeats)

(c) 0.9166 (it ends) (d) 0.91666… (only the 6 repeats)

Answer: (d) 0.91666… (only the 6 repeats)

How to see what this asks: Long division with a remainder that returns; see Type 4.

Given: The fraction 11/12.

Rule: Divide the top by the bottom. If a remainder comes back, the digits repeat from there.

Working: 110 ÷ 12 = 9 remainder 2. Then 20 ÷ 12 = 1 remainder 8. Then 80 ÷ 12 = 6 remainder 8. The remainder 8 has come back, so the digit 6 repeats forever.

Result: 0.91666…, with only the 6 repeating.

Check: 12 × 0.91666… = 10.9999… = 11 ✓.

Why the others are wrong: "0.9111… (only the 1 repeats)" stops the division one step too early. "0.9333… (only the 3 repeats)" has the wrong digits. "0.9166 (it ends)" is impossible because 12 has the prime factor 3.

Concept behind the question: *In 11/12 = 11/(2 × 2 × 3) the prime 3 forces a repeating decimal (Rule R4).*

Revise in the chapter document: Type 4 (Fractions to decimals: ending or repeating?) · Rule R4 · Example 11

Part 3 – Roots, irrational numbers and sums of natural numbers

A. Concept questions

Q61 · Part 3 · Concept · Level E · MCQ

Q61. Which of the following is an irrational number?

(a) √36 (b) √17

(c) √(4/9) (d) √121 ÷ 11

Answer: (b) √17

How to see what this asks: Simplify each option and see whether it becomes p/q; see Type 7.

Given: Four expressions with roots.

Rule: The root of a perfect square (or of a fraction of perfect squares) is rational; the root of a non-perfect-square is irrational.

Working: √36 = 6. √(4/9) = 2/3. √121 ÷ 11 = 11 ÷ 11 = 1. But 17 lies between 16 and 25 and is not a perfect square, so √17 is irrational.

Result: √17.

Check: 4 × 4 = 16 and 5 × 5 = 25, so no whole number squares to 17 ✓.

Why the others are wrong: "√36" equals 6. "√(4/9)" equals 2/3. "√121 ÷ 11" equals 1.

Concept behind the question: *A root sign alone does not make a number irrational; only roots of non-perfect-squares do.*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 29

Q62 · Part 3 · Concept · Level E · MCQ

Q62. Which of the following statements about π is true?

(a) π is exactly equal to 22/7 (b) π is exactly equal to 3.14

(c) π is an irrational number (d) π is a terminating decimal

Answer: (c) π is an irrational number

How to see what this asks: A fact about the number π; see Type 7.

Given: Four statements about π.

Rule: π = 3.14159265… never ends and never repeats, so π is irrational.

Working: 22/7 = 3.142857… and 3.14 are only approximations to π. A terminating decimal would be rational, which π is not.

Result: π is an irrational number.

Check: 22/7 = 3.142857… differs from 3.141592… from the third decimal place onwards ✓.

Why the others are wrong: "π is exactly equal to 22/7" 22/7 is only an approximation. "π is exactly equal to 3.14" 3.14 is only a rounded value. "π is a terminating decimal" its decimal never ends.

Concept behind the question: *π is irrational; 22/7 and 3.14 are convenient approximations.*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 25 · Rule R7 (π)

Q63 · Part 3 · Concept · Level E · MCQ

Q63. The product of a non-zero rational number and an irrational number is:

(a) always rational
(b) sometimes rational
(c) always an integer
(d) always irrational

Answer: (d) always irrational

How to see what this asks: A rule about combining rational and irrational numbers; see Type 7.

Given: A non-zero rational number r and an irrational number s.

Rule: Suppose r × s were rational. Then s = (r × s) ÷ r would also be rational, because dividing by a non-zero rational gives a rational.

Working: But s is irrational, so the assumption is false. Hence r × s is irrational. Example: 3 × √2 = 3√2 is irrational.

Result: Always irrational.

Check: Check with r = 1/2 and s = √2: (1/2) × √2 ≈ 0.7071…, which never ends or repeats ✓.

Why the others are wrong: "always rational" would force the irrational number to be rational. "sometimes rational" fails; this only happens when the rational number is 0. "always an integer" is false because irrational numbers are not integers.

Concept behind the question: *Non-zero rational × irrational = irrational (and so does rational + irrational).*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 26

Q64 · Part 3 · Concept · Level M · MCQ

Q64. Which of the following is a rational number?

(a) $(\sqrt{3})^2$
(b) $\sqrt{7} - 2$
(c) $\pi + 1$
(d) $4/\sqrt{3}$

Answer: (a) $(\sqrt{3})^2$

How to see what this asks: Simplify each expression; see Type 7.

Given: Four expressions.

Rule: $(\sqrt{n})^2 = n$; rational ± irrational is irrational; dividing by an irrational number gives an irrational number.

Working: $(\sqrt{3})^2 = 3$, a rational number. $\sqrt{7} - 2$ is irrational minus rational, so irrational. $\pi + 1$ is irrational plus rational, so irrational. $4/\sqrt{3} = 4\sqrt{3}/3$ is irrational.

Result: $(\sqrt{3})^2$.

Check: 1.732 × 1.732 ≈ 3.0 ✓.

Why the others are wrong: "$\sqrt{7} - 2$" irrational minus rational is irrational. "$\pi + 1$" irrational plus rational is irrational. "$4/\sqrt{3}$" equals $4\sqrt{3}/3$, which is irrational.

Concept behind the question: *Squaring a square root removes the root: $(\sqrt{n})^2 = n$.*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 and R9 · Example 35

Q65 · Part 3 · Concept · Level M · MCQ

Q65. How many of √4, √5, √9, √10 and √16 are irrational numbers?

(a) 2 (b) 3

(c) 1 (d) 5

Answer: (a) 2

How to see what this asks: Sort the roots into perfect squares and non-perfect-squares; see Type 7.

Given: Five square roots.

Rule: The root of a perfect square is a whole number (rational); the root of any other whole number is irrational.

Working: 4, 9 and 16 are perfect squares, so √4 = 2, √9 = 3 and √16 = 4 are rational. 5 and 10 are not perfect squares (4 < 5 < 9 and 9 < 10 < 16), so √5 and √10 are irrational.

Result: 2.

Check: 2 irrational + 3 rational = 5 numbers in all ✓.

Why the others are wrong: "3" counts one perfect-square root as irrational. "1" misses one of the irrational roots. "5" treats every root sign as irrational.

Concept behind the question: *Only roots of non-perfect-square whole numbers are irrational.*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 25

Q66 · Part 3 · Concept · Level E · MCQ

Q66. Which of the following numbers is irrational?

(a) 0.272727… (b) 0.2020020002… (one more 0 each time)

(c) 0.75 (d) 22/7

Answer: (b) 0.2020020002… (one more 0 each time)

How to see what this asks: An irrational number has a decimal that never ends and never repeats; see Type 7.

Given: Four numbers; exactly one is irrational.

Rule: A number is rational if it can be written as p/q, or if its decimal ends or repeats. Otherwise it is irrational.

Working: In 0.2020020002… the number of zeros between the 2s grows each time, so there is no repeating block and the decimal never ends. 0.272727… repeats the block 27. 0.75 = 3/4 ends. 22/7 is already p/q.

Result: 0.2020020002….

Check: The other three are all of the form p/q, so they cannot be irrational ✓.

Why the others are wrong: "0.272727…" repeats the block 27, so it is rational. "0.75" ends, so it is 3/4. "22/7" is p/q, so it is rational.

Concept behind the question: *Irrational: never ends and never repeats (Rule R7).*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 25

Q67 · Part 3 · Concept · Level E · MCQ

Q67. Which of the following is a perfect cube?

(a) 144 (b) 216

(c) 250 (d) 400

Answer: (b) 216

How to see what this asks: A perfect cube is a whole number times itself three times; see Type 6.

Given: Four whole numbers.

Rule: A perfect cube is a whole number multiplied by itself three times. Each prime in it appears in groups of three.

Working: 216 = 6 × 6 × 6, so it is a perfect cube. 144 = 12 × 12 is a perfect square, not a cube. 250 = 2 × 5 × 5 × 5 has a single 2, so the primes are not in threes. 400 = 20 × 20 = 2 × 2 × 2 × 2 × 5 × 5 is a perfect square.

Result: 216.

Check: 6 × 6 = 36 and 36 × 6 = 216 ✓.

Why the others are wrong: "144" is 12 × 12, a square but not a cube. "250" has the primes 2 × 5 × 5 × 5, and the single 2 spoils the groups. "400" is 20 × 20, a square but not a cube.

Concept behind the question: *A cube has every prime in groups of three (Section 1.2, Rule R6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 20 · Section 1.2 (Perfect square and perfect cube)

Q68 · Part 3 · Concept · Level E · MCQ

Q68. The sum of a rational number and an irrational number is:

(a) always rational
(b) sometimes rational
(c) always irrational
(d) always an integer

Answer: (c) always irrational

How to see what this asks: A rule about combining rational and irrational numbers; see Type 7.

Given: A rational number r and an irrational number s.

Rule: Rational + irrational is irrational.

Working: Suppose r + s were rational, say equal to t. Then s = t − r, and a rational number minus a rational number is rational. That would make s rational, which is not so. So r + s is irrational.

Result: Always irrational.

Check: For example 4 + √3 ≈ 5.732..., a decimal with no pattern ✓.

Why the others are wrong: "always rational" would make the irrational number rational. "sometimes rational" is false because no case works. "always an integer" is false because an integer is rational.

Concept behind the question: *Rational + irrational is irrational, by a short contradiction argument (Rule R7).*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 26

Q69 · Part 3 · Concept · Level M · MCQ

Q69. Which of the following statements about irrational numbers is true?

(a) The sum of two irrational numbers is always irrational.
(b) The product of two irrational numbers is always irrational.
(c) The square root of every whole number is irrational.
(d) The difference of two irrational numbers can be rational.

Answer: (d) The difference of two irrational numbers can be rational.

How to see what this asks: Look for one counter-example to each wrong statement; see Type 7.

Given: Four statements; only one is true.

Rule: Two irrational numbers can add, subtract or multiply to give a rational number. A root of a perfect square is a whole number.

Working: $(3 + \sqrt{2}) - \sqrt{2} = 3$, which is rational, and both $(3 + \sqrt{2})$ and $\sqrt{2}$ are irrational. So the first statement is true. For the second, $\sqrt{2} + (-\sqrt{2}) = 0$. For the third, $\sqrt{2} \times \sqrt{2} = 2$. For the fourth, $\sqrt{16} = 4$.

Result: The difference of two irrational numbers can be rational.

Check: One counter-example is enough to break each false statement ✓.

Why the others are wrong: "The sum of two irrational numbers is always irrational." fails for $\sqrt{2} + (-\sqrt{2}) = 0$. "The product of two irrational numbers is always irrational." fails for $\sqrt{2} \times \sqrt{2} = 2$. "The square root of every whole number is irrational." fails for $\sqrt{16} = 4$.

Concept behind the question: *Irrational numbers can cancel each other, so "always" statements about them are usually false (Rule R7).*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 27

Q70 · Part 3 · Concept · Level E · MCQ

Q70. A whole number is a perfect square only if, in its prime factorisation, every prime appears:

(a) an odd number of times
(b) three times
(c) an even number of times
(d) exactly once

Answer: (c) an even number of times

How to see what this asks: Think of how a square is made from pairs of equal primes; see Type 6.

Given: A whole number that is a perfect square.

Rule: A perfect square is a whole number times itself, so every prime in it comes in pairs.

Working: If $n = m \times m$, then every prime in m appears twice in n. For example $36 = 6 \times 6 = 2 \times 2 \times 3 \times 3$, so 2 appears twice and 3 appears twice. Pairs mean an even count.

Result: An even number of times.

Check: $36 = 2 \times 2 \times 3 \times 3$ has both primes twice ✓.

Why the others are wrong: "an odd number of times" would leave an unpaired prime, as in $12 = 2 \times 2 \times 3$. "three times" is the rule for a perfect cube. "exactly once" leaves every prime unpaired.

Concept behind the question: *Pairs of primes make a square; triples make a cube (Rule R6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 23

B. Calculation questions

Q71 · Part 3 · Calculation · Level E · MCQ

Q71. The value of $\sqrt{256}$ is:

(a) 14
(b) 16
(c) 18
(d) 128

Answer: (b) 16

How to see what this asks: A square root of a perfect square; see Type 6.

Given: √256.

Rule: √n is the number that gives n when multiplied by itself.

Working: Try 16: 16 × 16 = 256. (15 × 15 = 225 is too small and 17 × 17 = 289 is too big.)

Result: 16.

Check: 16 × 16 = 256 ✓.

Why the others are wrong: "14" gives 14 × 14 = 196. "18" gives 18 × 18 = 324. "128" is half of 256, not its square root.

Concept behind the question: *A square root undoes squaring.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 20

Q72 · Part 3 · Calculation · Level E · MCQ

Q72. The value of $\sqrt[3]{729}$ is:

(a) 27 (b) 81

(c) 243 (d) 9

Answer: (d) 9

How to see what this asks: A cube root of a perfect cube; see Type 6.

Given: $\sqrt[3]{729}$.

Rule: $\sqrt[3]{n}$ is the number that gives n when multiplied by itself three times.

Working: Try 9: 9 × 9 = 81 and 81 × 9 = 729.

Result: 9.

Check: 9 × 9 × 9 = 729 ✓.

Why the others are wrong: "27" is the square root of 729, not the cube root. "81" is 729 ÷ 9. "243" is 729 ÷ 3.

Concept behind the question: *A cube root undoes cubing; do not confuse it with a square root.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 20

Q73 · Part 3 · Calculation · Level E · MCQ

Q73. The value of √0.81 is:

(a) 0.09 (b) 0.45

(c) 0.9 (d) 9

Answer: (c) 0.9

How to see what this asks: A square root of a decimal; see Type 6.

Given: √0.81.

Rule: Write the decimal as a fraction over a power of 10: 0.81 = 81/100.

Working: √81 = 9 and √100 = 10, so √0.81 = 9/10 = 0.9.

Result: 0.9.

Check: 0.9 × 0.9 = 0.81 ✓.

Why the others are wrong: "0.09" copies the digits of the root without placing the decimal point correctly. "0.45" halves 0.81 instead of taking the root. "9" ignores the decimal point.

Concept behind the question: *The root of a decimal has half as many decimal places.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 22

Q74 · Part 3 · Calculation · Level M · MCQ

Q74. √91 lies between:

(a) 8 and 9 (b) 9 and 10

(c) 10 and 11 (d) 90 and 92

Answer: (b) 9 and 10

How to see what this asks: Trap the root between two perfect squares; see Type 6.

Given: √91, and 91 is not a perfect square.

Rule: Find the perfect squares just below and just above 91 and take their roots.

Working: 9 × 9 = 81 and 10 × 10 = 100, and 81 < 91 < 100. So 9 < √91 < 10.

Result: 9 and 10.

Check: 9.5 × 9.5 = 90.25, a little less than 91, so √91 is just above 9.5 and below 10 ✓.

Why the others are wrong: "8 and 9" would need 91 to be between 64 and 81. "10 and 11" would need 91 to be between 100 and 121. "90 and 92" compares the number itself, not its root.

Concept behind the question: *A root lies between the roots of the two nearest perfect squares.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 21

Q75 · Part 3 · Calculation · Level E · MCQ

Q75. The value of √(18 × 8) is:

(a) 13 (b) 144

(c) 12 (d) 26

Answer: (c) 12

How to see what this asks: Multiply first, then take the root; see Type 6.

Given: √(18 × 8).

Rule: Multiply under the root sign, then take the root.

Working: 18 × 8 = 144 and √144 = 12, since 12 × 12 = 144.

Result: 12.

Check: √18 × √8 ≈ 4.243 × 2.828 ≈ 12.0 ✓.

Why the others are wrong: "13" averages 18 and 8. "144" forgets to take the root. "26" adds 18 and 8.

Concept behind the question: *The product under a root can turn into a perfect square: √(a × b) = √a × √b.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 27 · Rule R6 (√(a × b) = √a × √b)

Q76 · Part 3 · Calculation · Level M · MCQ

Q76. The smallest number by which 450 must be multiplied to get a perfect square is:

(a) 3 (b) 2

(c) 5 (d) 6

Answer: (b) 2

How to see what this asks: Look for the unpaired prime in the factorisation; see Type 6.

Given: 450.

Rule: A number is a perfect square when every prime appears an even number of times.

Working: 450 ÷ 2 = 225, 225 ÷ 3 = 75, 75 ÷ 3 = 25, 25 ÷ 5 = 5, 5 ÷ 5 = 1, so 450 = 2 × 3 × 3 × 5 × 5. The 3s and the 5s are in pairs but the 2 is alone, so we multiply by 2.

Result: 2.

Check: 450 × 2 = 900 = 30 × 30 ✓.

Why the others are wrong: "3" gives three 3s, which cannot all be paired. "5" gives three 5s, which cannot all be paired. "6" also breaks the pairs of 3s.

Concept behind the question: *Multiply by the unpaired primes, and only those.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 23

Q77 · Part 3 · Calculation · Level H · MCQ

Q77. The smallest number by which 3087 must be divided to get a perfect cube is:

(a) 3 (b) 7

(c) 9 (d) 63

Answer: (c) 9

How to see what this asks: Group the primes in threes and divide away what is left over; see Type 6.

Given: 3087.

Rule: A number is a perfect cube when every prime appears in groups of three.

Working: 3087 ÷ 3 = 1029, 1029 ÷ 3 = 343, 343 ÷ 7 = 49, 49 ÷ 7 = 7, 7 ÷ 7 = 1, so 3087 = 3 × 3 × 7 × 7 × 7. The three 7s form a complete group, but the two 3s are only two, not three. Dividing by 3 × 3 = 9 leaves 7 × 7 × 7 = 343.

Result: 9.

Check: 3087 ÷ 9 = 343 = 7 × 7 × 7 ✓.

Why the others are wrong: "3" leaves $1029 = 3 \times 7^3$, which still has a single 3. "7" leaves $441 = 3^2 \times 7^2$, which is not a cube. "63" leaves 49, which is not a cube.

Concept behind the question: *For a cube, divide out the primes that are left over after forming groups of three.*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 24

Q78 · Part 3 · Calculation · Level E · MCQ

Q78. The value of $(3 + \sqrt{2})(3 - \sqrt{2})$ is:

(a) 11 (b) 9

(c) 1 (d) 7

Answer: (d) 7

How to see what this asks: A product of conjugate brackets; see Type 9.

Given: $(3 + \sqrt{2})(3 - \sqrt{2})$.

Rule: $(a + b)(a - b) = a^2 - b^2$.

Working: Here a = 3 and b = √2, so $3^2 - (\sqrt{2})^2 = 9 - 2 = 7$.

Result: 7.

Check: (3 + 1.414)(3 − 1.414) = 4.414 × 1.586 ≈ 7.0 ✓.

Why the others are wrong: "11" adds the squares: 9 + 2. "9" forgets to subtract $(\sqrt{2})^2$. "1" subtracts without squaring: 3 − 2.

Concept behind the question: *Conjugate brackets multiply to a rational number: the roots cancel.*

Revise in the chapter document: Type 9 (Roots inside brackets) · Rule R9 · Example 35

Q79 · Part 3 · Calculation · Level M · MCQ

Q79. The value of $(5 - \sqrt{2})^2$ is:

(a) $27 + 10\sqrt{2}$ (b) $27 - 10\sqrt{2}$

(c) 27 (d) 23

Answer: (b) $27 - 10\sqrt{2}$

How to see what this asks: Squaring a bracket with a root; see Type 9.

Given: $(5 - \sqrt{2})^2$.

Rule: $(a - b)^2 = a^2 - 2ab + b^2$.

Working: Here $a = 5$ and $b = \sqrt{2}$. So $a^2 = 25$, $2ab = 2 \times 5 \times \sqrt{2} = 10\sqrt{2}$ and $b^2 = 2$. The result is $25 - 10\sqrt{2} + 2 = 27 - 10\sqrt{2}$.

Result: $27 - 10\sqrt{2}$.

Check: $5 - 1.4142 = 3.5858$, and $3.5858^2 \approx 12.858$. Also $27 - 10 \times 1.4142 = 12.858$ ✓.

Why the others are wrong: "$27 + 10\sqrt{2}$" gets the sign of the middle term wrong. "27" forgets the middle term. "23" uses $a^2 - b^2$ instead of the square of the bracket.

Concept behind the question: *The middle term 2ab is what keeps the answer irrational.*

Revise in the chapter document: Type 9 (Roots inside brackets) · Rule R9 · Example 36

Q80 · Part 3 · Calculation · Level M · MCQ

Q80. Which of the following is an irrational number that lies between 5 and 6?

(a) $\sqrt{30}$ (b) 5.5

(c) $\sqrt{25}$ (d) $\sqrt{37}$

Answer: (a) $\sqrt{30}$

How to see what this asks: Write 5 and 6 as $\sqrt{25}$ and $\sqrt{36}$, then pick a non-square between 25 and 36; see Type 7.

Given: We need a number that is irrational and also between 5 and 6.

Rule: $5 = \sqrt{25}$ and $6 = \sqrt{36}$. The root of a whole number between 25 and 36 that is not a perfect square is irrational and lies between 5 and 6.

Working: $\sqrt{30}$: $25 < 30 < 36$, so $5 < \sqrt{30} < 6$, and 30 is not a perfect square, so $\sqrt{30}$ is irrational. 5.5 lies between 5 and 6 but it is 11/2, which is rational. $\sqrt{25} = 5$, which is not between 5 and 6 and is rational. $\sqrt{37}$: 37 is above 36, so $\sqrt{37}$ is more than 6.

Result: $\sqrt{30}$.

Check: $5.4^2 = 29.16$ and $5.5^2 = 30.25$, so $\sqrt{30}$ is about 5.48, which is between 5 and 6 ✓.

Why the others are wrong: "5.5" is between 5 and 6 but is rational. "$\sqrt{25}$" equals 5, which is rational and not between 5 and 6. "$\sqrt{37}$" is about 6.08, which is more than 6.

Concept behind the question: *To place an irrational number between two whole numbers, square them and pick a non-square in between.*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R7 · Example 28

Q81 · Part 3 · Calculation · Level E · MCQ

Q81. The value of $1 + 2 + 3 + \ldots + 50$ is:

(a) 1250 (b) 1275

(c) 2550 (d) 1225

Answer: (b) 1275

How to see what this asks: The sum of the first n natural numbers; see Type 8.

Given: The sum of the first 50 natural numbers.

Rule: $1 + 2 + \ldots + n = n(n + 1) \div 2$.

Working: n = 50, so $50 \times 51 \div 2 = 2550 \div 2 = 1275$.

Result: 1275.

Check: Pairing: 25 pairs, each adding to 51, and $25 \times 51 = 1275$ ✓.

Why the others are wrong: "1250" is $50 \times 50 \div 2$, using n instead of n + 1. "2550" forgets to divide by 2. "1225" is $49 \times 50 \div 2$, using n − 1 instead of n + 1.

Concept behind the question: *The sum of the first n natural numbers is $n(n + 1) \div 2$.*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 30

Q82 · Part 3 · Calculation · Level E · MCQ

Q82. The sum of the first 14 odd numbers is:

(a) 210 (b) 182

(c) 28 (d) 196

Answer: (d) 196

How to see what this asks: The sum of the first n odd numbers; see Type 8.

Given: The first 14 odd numbers: 1, 3, 5, ..., 27.

Rule: The sum of the first n odd numbers is n^2.

Working: $14^2 = 14 \times 14 = 196$.

Result: 196.

Check: Small case: the first 3 odd numbers give $1 + 3 + 5 = 9 = 3^2$ ✓.

Why the others are wrong: "210" uses the even-number rule $n(n + 1) = 14 \times 15$. "182" uses 13×14. "28" is 2×14 (the last odd number plus 1), not the sum.

Concept behind the question: *Odd numbers add up to a perfect square: the first n odd numbers sum to n^2.*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 31

Q83 · Part 3 · Calculation · Level E · MCQ

Q83. The sum of the first 16 even numbers is:

(a) 256 (b) 272

(c) 136 (d) 240

Answer: (b) 272

How to see what this asks: The sum of the first n even numbers; see Type 8.

Given: The first 16 even numbers: 2, 4, 6, ..., 32.

Rule: The sum of the first n even numbers is $n(n + 1)$.

Working: $16 \times 17 = 272$.

Result: 272.

Check: It is twice the sum of 1 to 16. That sum is $16 \times 17 \div 2 = 136$, and $2 \times 136 = 272$ ✓.

Why the others are wrong: "256" uses the odd-number rule n^2. "136" is the sum of 1 to 16, not of the even numbers. "240" is 15×16.

Concept behind the question: *The even numbers add up to $n(n + 1)$, double the sum of the first n natural numbers.*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 31

Q84 · Part 3 · Calculation · Level M · MCQ

Q84. The sum of seven consecutive integers is 112. The smallest of them is:

(a) 16 (b) 19
(c) 12 (d) 13

Answer: (d) 13

How to see what this asks: Write the integers around the middle one; see Type 8.

Given: Seven consecutive integers with sum 112.

Rule: For an odd count, the middle integer is the sum divided by the count. The others lie 1, 2 and 3 steps on each side of it.

Working: The middle integer is 112 ÷ 7 = 16. The integers are 13, 14, 15, 16, 17, 18, 19.

Result: The smallest is 13.

Check: 13 + 14 + 15 + 16 + 17 + 18 + 19 = 112 ✓.

Why the others are wrong: "16" is the middle integer, not the smallest. "19" is the largest integer. "12" comes from stepping back 4 from the middle, but only 3 steps are needed.

Concept behind the question: *For an odd count of consecutive integers, the middle one is the average; count back to find the smallest.*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 32

Q85 · Part 3 · Calculation · Level H · MCQ

Q85. The sum of all the integers from 21 to 40 (both included) is:

(a) 610 (b) 600
(c) 630 (d) 589

Answer: (a) 610

How to see what this asks: Subtract one sum of natural numbers from another; see Type 8.

Given: 21 + 22 + … + 40.

Rule: 1 + 2 + … + n = n(n + 1) ÷ 2. The required sum = (1 to 40) − (1 to 20).

Working: 1 to 40: 40 × 41 ÷ 2 = 820. 1 to 20: 20 × 21 ÷ 2 = 210. So the sum = 820 − 210 = 610.

Result: 610.

Check: There are 20 numbers with average (21 + 40) ÷ 2 = 30.5, and 20 × 30.5 = 610 ✓.

Why the others are wrong: "600" uses the average 30 instead of 30.5. "630" is the sum from 20 to 40, one number too many. "589" is the sum from 22 to 40, one number too few.

Concept behind the question: *A sum that does not start at 1 is the difference of two sums that do.*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 34

Q86 · Part 3 · Calculation · Level E · MCQ

Q86. The value of √361 is:

(a) 19 (b) 21
(c) 18 (d) 17

Answer: (a) 19

How to see what this asks: A square root of a perfect square; see Type 6.

Given: √361.

Rule: The square root is the number that gives 361 when multiplied by itself.

Working: Try numbers near the answer. 18 × 18 = 324 is too small and 20 × 20 = 400 is too big. 19 × 19 = 361.

Result: 19.

Check: 19 × 19 = 361 ✓.

Why the others are wrong: "21" gives 21 × 21 = 441. "18" gives 18 × 18 = 324. "17" gives 17 × 17 = 289.

Concept behind the question: *A square root undoes squaring (Section 1.2, Type 6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 20

Q87 · Part 3 · Calculation · Level E · MCQ

Q87. The value of $\sqrt[3]{512}$ is:

(a) 9 (b) 7

(c) 8 (d) 171

Answer: (c) 8

How to see what this asks: A cube root of a perfect cube; see Type 6.

Given: $\sqrt[3]{512}$.

Rule: The cube root is the number that gives 512 when multiplied by itself three times.

Working: 512 = 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2, which is nine 2s. Group them in threes: (2, 2, 2), (2, 2, 2), (2, 2, 2). One 2 from each group gives 2 × 2 × 2 = 8.

Result: 8.

Check: 8 × 8 = 64 and 64 × 8 = 512 ✓.

Why the others are wrong: "9" gives 9 × 9 × 9 = 729. "7" gives 7 × 7 × 7 = 343. "171" is 512 ÷ 3, which divides by 3 instead of taking a cube root.

Concept behind the question: *Groups of three equal primes give a cube root (Rule R6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 20

Q88 · Part 3 · Calculation · Level E · MCQ

Q88. The value of √(121/225) is:

(a) 11/225 (b) 11/15

(c) 121/15 (d) 12/15

Answer: (b) 11/15

How to see what this asks: Take the root of the top and the root of the bottom; see Type 6.

Given: √(121/225).

Rule: √(a/b) = √a ÷ √b.

Working: √121 = 11, because 11 × 11 = 121. √225 = 15, because 15 × 15 = 225. So the value is 11/15.

Result: 11/15.

Check: $(11/15)^2 = 121/225$ ✓.

Why the others are wrong: "11/225" takes the root of the top only. "121/15" takes the root of the bottom only. "12/15" takes 12 as the root of 121, but 12 × 12 = 144.

Concept behind the question: *Take the root of the top and of the bottom separately (Rule R6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 22

Q89 · Part 3 · Calculation · Level M · MCQ

Q89. √130 lies between:

(a) 11 and 12 (b) 10 and 11

(c) 12 and 13 (d) 13 and 14

Answer: (a) 11 and 12

How to see what this asks: Trap the number between two perfect squares; see Type 6.

Given: The number 130, which is not a perfect square.

Rule: Find the perfect squares just below and just above 130. Their roots are the two whole numbers.

Working: $11 \times 11 = 121$ and $12 \times 12 = 144$, and $121 < 130 < 144$. Taking square roots gives $11 < \sqrt{130} < 12$.

Result: 11 and 12.

Check: $11.4 \times 11.4 = 129.96$, which is just below 130 ✓.

Why the others are wrong: "10 and 11" has squares 100 and 121, both below 130. "12 and 13" has squares 144 and 169, both above 130. "13 and 14" has squares 169 and 196, both above 130.

Concept behind the question: *A root lies between the roots of the two nearest perfect squares (Type 6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 21

Q90 · Part 3 · Calculation · Level M · MCQ

Q90. If 1008 is multiplied by the smallest possible number so that the product is a perfect square, the square root of the product is:

(a) 42 (b) 84
(c) 7 (d) 7056

Answer: (b) 84

How to see what this asks: Make the product a perfect square first, then take its root; see Type 6.

Given: The number 1008.

Rule: A perfect square has every prime in pairs. Multiply by each prime that is left without a partner. Then the square root takes one prime from each pair.

Working: $1008 \div 2 = 504$, $504 \div 2 = 252$, $252 \div 2 = 126$, $126 \div 2 = 63$, $63 \div 3 = 21$, $21 \div 3 = 7$, $7 \div 7 = 1$. So $1008 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7$. The 7 is alone, so multiply by 7. The product is $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 7$. One prime from each pair gives $2 \times 2 \times 3 \times 7 = 84$.

Result: 84.

Check: $84 \times 84 = 7056$ and $1008 \times 7 = 7056$ ✓.

Why the others are wrong: "42" takes only half of the correct root. "7" is the multiplier, not the square root. "7056" is the product itself, not its square root.

Concept behind the question: *First find the missing partner prime, then take one prime from each pair for the root (Rule R6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 23

Q91 · Part 3 · Calculation · Level M · MCQ

Q91. The smallest number by which 1176 must be divided so that the quotient is a perfect square is:

(a) 2 (b) 6
(c) 3 (d) 12

Answer: (b) 6

How to see what this asks: Divide out the primes that are left without a partner; see Type 6.

Given: The number 1176.

Rule: A perfect square has every prime in pairs. Divide out the primes that have no partner.

Working: 1176 ÷ 2 = 588, 588 ÷ 2 = 294, 294 ÷ 2 = 147, 147 ÷ 3 = 49, 49 ÷ 7 = 7, 7 ÷ 7 = 1. So 1176 = 2 × 2 × 2 × 3 × 7 × 7. The 7s make a pair. Two of the 2s make a pair, but one 2 is alone, and the single 3 is alone. Divide by 2 × 3 = 6.

Result: 6.

Check: 1176 ÷ 6 = 196 and 14 × 14 = 196 ✓.

Why the others are wrong: "2" leaves 588 = 2 × 2 × 3 × 7 × 7, and the 3 is still alone. "3" leaves 392 = 2 × 2 × 2 × 7 × 7, and a 2 is still alone. "12" leaves 98 = 2 × 7 × 7, so a 2 is now unpaired.

Concept behind the question: *Dividing out the unpaired primes leaves a perfect square (Rule R6; the same idea is used for cubes in Example 24).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 23 · Example 24 (dividing out)

Q92 · Part 3 · Calculation · Level M · MCQ

Q92. The value of √0.0289 is:

(a) 0.17 (b) 0.017

(c) 1.7 (d) 0.0017

Answer: (a) 0.17

How to see what this asks: A square root of a decimal; see Type 6.

Given: √0.0289.

Rule: Write the decimal as a whole number over 10, 100, 1000, ... Then take the root of the top and the bottom.

Working: 0.0289 has four digits after the point, so 0.0289 = 289/10000. Now 17 × 17 = 289 and 100 × 100 = 10000. So the root is 17/100 = 0.17.

Result: 0.17.

Check: 0.17 × 0.17 = 0.0289 ✓.

Why the others are wrong: "0.017" has three places, but the root needs half of four places, which is two. "1.7" puts the point in the wrong place. "0.0017" has four places, but the root has half as many places as 0.0289.

Concept behind the question: *The root of a decimal has half as many decimal places (Type 6, Example 22).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 22

Q93 · Part 3 · Calculation · Level M · MCQ

Q93. The value of $\sqrt[3]{2744}$ is:

(a) 12 (b) 14

(c) 16 (d) 13

Answer: (b) 14

How to see what this asks: Group the primes of 2744 in threes; see Type 6.

Given: $\sqrt[3]{2744}$.

Rule: Factorise the number and group equal primes in threes. Take one prime from each group.

Working: 2744 ÷ 2 = 1372, 1372 ÷ 2 = 686, 686 ÷ 2 = 343, 343 ÷ 7 = 49, 49 ÷ 7 = 7, 7 ÷ 7 = 1. So 2744 = 2 × 2 × 2 × 7 × 7 × 7. The groups are (2, 2, 2) and (7, 7, 7). The cube root is 2 × 7 = 14.

Result: 14.

Check: 14 × 14 = 196 and 196 × 14 = 2744 ✓.

Why the others are wrong: "12" gives 12 × 12 × 12 = 1728. "16" gives 16 × 16 × 16 = 4096. "13" gives 13 × 13 × 13 = 2197.

Concept behind the question: *Triples of equal primes give a cube root (Rule R6).*

Revise in the chapter document: Type 6 (Squares, cubes and their roots) · Rule R6 · Example 20

Q94 · Part 3 · Calculation · Level M · MCQ

Q94. Which of the following expressions is irrational?

(a) $\sqrt{12} \times \sqrt{3}$ (b) $\sqrt{18} \div \sqrt{2}$

(c) $(\sqrt{7})^2$ (d) $\sqrt{5} + \sqrt{5}$

Answer: (d) $\sqrt{5} + \sqrt{5}$

How to see what this asks: Simplify each option; see Type 7.

Given: Four expressions made of square roots; exactly one is irrational.

Rule: $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$ and $\sqrt{a} \div \sqrt{b} = \sqrt{(a/b)}$. A root times itself gives the number under the root.

Working: $\sqrt{5} + \sqrt{5} = 2\sqrt{5}$, a non-zero rational times an irrational number, so it is irrational. $\sqrt{12} \times \sqrt{3} = \sqrt{36} = 6$. $\sqrt{18} \div \sqrt{2} = \sqrt{9} = 3$. $(\sqrt{7})^2 = 7$.

Result: $\sqrt{5} + \sqrt{5} = 2\sqrt{5}$.

Check: $2\sqrt{5} \approx 4.4721\ldots$, with no pattern, while the other three are 6, 3 and 7 ✓.

Why the others are wrong: "$\sqrt{12} \times \sqrt{3}$" equals $\sqrt{36} = 6$, which is rational. "$\sqrt{18} \div \sqrt{2}$" equals $\sqrt{9} = 3$, which is rational. "$(\sqrt{7})^2$" equals 7, which is rational.

Concept behind the question: *Combine the roots first; only the option that keeps a root is irrational (Rules R6 and R7).*

Revise in the chapter document: Type 7 (Irrational numbers) · Rule R6 and R7 · Example 29

Q95 · Part 3 · Calculation · Level E · MCQ

Q95. The value of $(8 + \sqrt{6})(8 - \sqrt{6})$ is:

(a) 70 (b) 64

(c) 2 (d) 58

Answer: (d) 58

How to see what this asks: A bracket times its partner; see Type 9.

Given: $(8 + \sqrt{6})(8 - \sqrt{6})$.

Rule: $(a + b)(a - b) = a^2 - b^2$. The middle terms cancel.

Working: Here $a = 8$ and $b = \sqrt{6}$. So $a^2 = 64$ and $b^2 = (\sqrt{6})^2 = 6$. The answer is $64 - 6 = 58$.

Result: 58.

Check: 8 + 2.4495 = 10.4495 and 8 − 2.4495 = 5.5505, and 10.4495 × 5.5505 ≈ 58.0 ✓.

Why the others are wrong: "70" adds the squares: 64 + 6. "64" forgets to subtract $(\sqrt{6})^2$. "2" subtracts without squaring: 8 − 6.

Concept behind the question: *The root parts cancel, so a whole number is left (Rule R9).*

Revise in the chapter document: Type 9 (Roots inside brackets) · Rule R9 · Example 35

Q96 · Part 3 · Calculation · Level M · MCQ

Q96. The value of $(\sqrt{7} + \sqrt{2})^2$ is:

(a) 9 (b) $9 + 2\sqrt{14}$

(c) $9 + \sqrt{14}$ (d) $9 - 2\sqrt{14}$

Answer: (b) $9 + 2\sqrt{14}$

How to see what this asks: Squaring a bracket with two roots; see Type 9.

Given: $(\sqrt{7} + \sqrt{2})^2$.

Rule: $(a + b)^2 = a^2 + 2ab + b^2$. Also $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$.

Working: $a^2 = (\sqrt{7})^2 = 7$ and $b^2 = (\sqrt{2})^2 = 2$. The middle term is $2 \times \sqrt{7} \times \sqrt{2} = 2\sqrt{14}$, because $\sqrt{7} \times \sqrt{2} = \sqrt{14}$. So the answer is $7 + 2\sqrt{14} + 2 = 9 + 2\sqrt{14}$.

Result: $9 + 2\sqrt{14}$.

Check: $\sqrt{7} + \sqrt{2} \approx 4.0600$ and its square is about 16.484. Also $9 + 2 \times 3.7417 = 16.483$ ✓.

Why the others are wrong: "9" forgets the middle term. "$9 + \sqrt{14}$" forgets the 2 in the middle term 2ab. "$9 - 2\sqrt{14}$" uses the sign of $(a - b)^2$.

Concept behind the question: *The middle term 2ab keeps a root, so the answer is irrational (Rule R9).*

Revise in the chapter document: Type 9 (Roots inside brackets) · Rule R9 · Example 36

Q97 · Part 3 · Calculation · Level M · MCQ

Q97. The value of $(5 + \sqrt{7})(5 - \sqrt{7}) + (\sqrt{7})^2$ is:

(a) 25 (b) 18

(c) 11 (d) 32

Answer: (a) 25

How to see what this asks: Two ideas together: a bracket times its partner, and a root times itself; see Type 9.

Given: $(5 + \sqrt{7})(5 - \sqrt{7}) + (\sqrt{7})^2$.

Rule: $(a + b)(a - b) = a^2 - b^2$, and $(\sqrt{b})^2 = b$.

Working: The first part is $5^2 - (\sqrt{7})^2 = 25 - 7 = 18$. The second part is $(\sqrt{7})^2 = 7$. Add them: $18 + 7 = 25$.

Result: 25.

Check: $18 + 7 = 25$, and the roots have left no irrational part ✓.

Why the others are wrong: "18" leaves out the second part. "11" subtracts the second part: $18 - 7$. "32" forgets to subtract $(\sqrt{7})^2$ in the first part, so it gets 25, then adds 7.

Concept behind the question: *Work out each part separately, then combine (Rule R9).*

Revise in the chapter document: Type 9 (Roots inside brackets) · Rule R9 · Example 35

Q98 · Part 3 · Calculation · Level E · MCQ

Q98. The sum of the first n odd numbers is 169. Then n is:

(a) 12 (b) 14

(c) 13 (d) 84

Answer: (c) 13

How to see what this asks: The sum of the first n odd numbers is $n \times n$; see Type 8.

Given: $1 + 3 + 5 + \ldots$ (n terms) = 169.

Rule: The sum of the first n odd numbers is n^2.

Working: $n^2 = 169$. Since $13 \times 13 = 169$, $n = 13$.

Result: 13.

Check: $13^2 = 169$ ✓.

Why the others are wrong: "12" gives $12^2 = 144$. "14" gives $14^2 = 196$. "84" is $169 \div 2$, which halves the sum instead of taking a square root.

Concept behind the question: *Odd numbers add up to a square: the first n of them sum to n^2 (Rule R8).*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 31

Q99 · Part 3 · Calculation · Level H · MCQ

Q99. The sum of the first 20 odd numbers minus the sum of the first 20 even numbers is:

(a) 20 (b) −40

(c) 0 (d) −20

Answer: (d) −20

How to see what this asks: Use both sum formulas and subtract; see Type 8.

Given: n = 20 for both sums.

Rule: The sum of the first n odd numbers is n^2. The sum of the first n even numbers is n(n + 1).

Working: Odd: 20 × 20 = 400. Even: 20 × 21 = 420. The difference is 400 − 420 = −20.

Result: −20.

Check: Each even number is 1 more than the odd number in the same position. So over 20 pairs the even sum is 20 more and the odd sum is smaller ✓.

Why the others are wrong: "20" has the right size and the wrong sign. "−40" uses n(n + 2) = 440 for the even sum. "0" assumes the two sums are equal.

Concept behind the question: *Odd numbers: n^2. Even numbers: n(n + 1). The even sum is always n bigger (Rule R8).*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 31

Q100 · Part 3 · Calculation · Level H · MCQ

Q100. The sum of all the integers from 51 to 80 (both included) is:

(a) 1914 (b) 1965

(c) 3240 (d) 2015

Answer: (b) 1965

How to see what this asks: A sum that does not start at 1 is one sum minus another; see Type 8.

Given: The numbers 51, 52, ..., 80.

Rule: Use n(n + 1) ÷ 2 for a sum from 1. Then sum from a to b = (sum from 1 to b) − (sum from 1 to a − 1).

Working: Sum from 1 to 80 = 80 × 81 ÷ 2 = 3240. Sum from 1 to 50 = 50 × 51 ÷ 2 = 1275. So the required sum is 3240 − 1275 = 1965.

Result: 1965.

Check: There are 80 − 51 + 1 = 30 numbers. The average of the first and last is (51 + 80) ÷ 2 = 65.5. Then 30 × 65.5 = 1965 ✓.

Why the others are wrong: "1914" takes away the sum up to 51 (1326), so the number 51 is lost. "3240" is the sum from 1 to 80 with nothing taken away. "2015" takes away the sum up to 49 (1225), so the number 50 is wrongly kept.

Concept behind the question: *For a sum that starts after 1, subtract the sum up to the number just before the start (Rule R8).*

Revise in the chapter document: Type 8 (Sums and patterns of natural numbers) · Rule R8 · Example 34

Answer key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	(c)	21	(a)	41	(c)	61	(b)	81	(b)
2	(c)	22	(b)	42	(a)	62	(c)	82	(d)
3	(a)	23	(d)	43	(b)	63	(d)	83	(b)
4	(b)	24	(a)	44	(a)	64	(a)	84	(d)
5	(a)	25	(c)	45	(d)	65	(a)	85	(a)
6	(d)	26	(c)	46	(d)	66	(b)	86	(a)
7	(a)	27	(d)	47	(c)	67	(b)	87	(c)
8	(c)	28	(a)	48	(a)	68	(c)	88	(b)
9	(b)	29	(b)	49	(d)	69	(d)	89	(a)
10	(d)	30	(a)	50	(a)	70	(c)	90	(b)
11	(c)	31	(a)	51	(c)	71	(b)	91	(b)
12	(c)	32	(c)	52	(b)	72	(d)	92	(a)
13	(d)	33	(a)	53	(d)	73	(c)	93	(b)
14	(b)	34	(c)	54	(b)	74	(b)	94	(d)
15	(c)	35	(a)	55	(d)	75	(c)	95	(d)
16	(d)	36	(b)	56	(a)	76	(b)	96	(b)
17	(c)	37	(c)	57	(b)	77	(c)	97	(a)
18	(c)	38	(d)	58	(c)	78	(d)	98	(c)
19	(d)	39	(a)	59	(a)	79	(b)	99	(d)
20	(d)	40	(d)	60	(d)	80	(a)	100	(b)